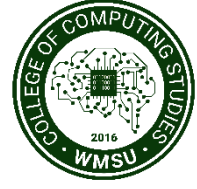




Republic of the Philippines
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College of Computing Studies

DEPARTMENT OF INFORMATION TECHNOLOGY

Zamboanga City

**CRIMSON MAP: A Hybrid Augmented Reality Indoor-Outdoor Navigation
System for University Campuses**

A Capstone Project Presented to the Faculty of
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CHAPTER I

INTRODUCTION

Background of the Study

Western Mindanao State University (WMSU) is one of the state universities located in Zamboanga City, established in 1904, and is one of the largest and most established institutions in Western Mindanao [18]. The university operates two major campuses: the Main Campus in Normal Road, Baliwasan, which covers approximately 79,000 square meters, and the San Ramon Campus, which spans around 200,000 square meters [18], [19]. WMSU comprises more than 15 colleges, including the College of Nursing, College of Engineering and Technology, College of Education, College of Social Work, and several other academic and administrative units [18], [19]. It serves a student population of over 32,000 and employs more than 600 faculty members, along with nearly 200 administrative staff [18], [19].

Given the university's large-scale and distributed campus layout, navigating the grounds can be overwhelming, especially for freshmen, transferees, parents, and visitors. Key facilities are often separated by considerable walking distances, with limited directional signages and no centralized digital wayfinding system in place. Students often rely on verbal directions and paper maps to find specific buildings or offices, leading to confusion, delays, and missed appointments or classes.

In recent years, modern technologies have provided new approaches to solving wayfinding problems in large and complex environments. Tools such as mobile GPS applications, indoor mapping systems, and location-based services have become common in places like malls, airports, and universities abroad. One of the most promising technologies for enhancing physical navigation is Augmented Reality (AR), which allows

digital navigation markers on a real-world view using a smartphone camera. This immersive approach to navigation opens new opportunities for institutions like WMSU to improve accessibility and convenience for all campus users through digital transformation.

Statement of the Problem

Navigating the Western Mindanao State University (WMSU) Main Campus is often difficult for new students, visitors, and guests due to its large and divided layout across three campuses—Campus A and Campus B in Baliwasan, and Campus C in San Ramon. The presence of numerous buildings and limited directional signage, combined with the lack of a unified digital wayfinding system, forces users to rely on verbal directions or outdated physical maps, leading to confusion, delays, and stress. Existing GPS-based tools such as Google Maps do not effectively support campus navigation since they focus on road networks rather than walkable pathways and lack real-time, intuitive guidance. Additionally, the absence of smart route planning and turn-by-turn navigation results in inefficient travel within the campus. These challenges emphasize the need for a modern, augmented reality–based navigation solution to improve accessibility and orientation within the WMSU Main Campus.

Objectives

To develop an Augmented Reality (AR)–based campus navigation mobile application for the Main Campus of Western Mindanao State University (WMSU) that enables users—including freshmen, visitors, and staff—to efficiently locate buildings and offices.

Specific Objectives

- Design and develop a mobile application using Unity and ARCore to support AR-based navigation, providing outdoor visual guidance with route updates.
- Implement GPS-based navigation as the primary localization method for outdoor routes, ensuring accurate positioning and tracking under normal GPS conditions.
- Provide audio-based navigational prompts to guide users during route navigation and improve usability and accessibility of the application.
- Implement a pathfinding algorithm that calculates and displays route between the user's current location and the selected destination.
- Develop a recalibration feature that allows users to scan QR codes only when GPS signals are weak or positional errors occur, restoring accurate location data for continued navigation.
- To allow users to view building layouts and key rooms through an indoor map accessible via an Indoor button when a floor plan is available.
- Develop a web-based admin panel for authorized campus staff to add, update, and connect buildings with distances and indoor layouts, and synchronize changes to the mobile app for updates.

Scope and Limitations

Scope

This project involves the development of a mobile application that provides Augmented Reality based outdoor navigation which comprises Campus A, Campus B, and Campus C. Campuses A and B are physically connected, allowing users to navigate between them. Campus C, however, is located separately and cannot be navigated to or from Campuses A or B.

The application is designed primarily to assist freshmen, visitors, and staff in locating buildings and offices using real-time AR markers and directional overlays.

The app provides multiple route options from the selected starting point to the destination, automatically highlighting the shortest route as “Recommended.” Path calculations are handled using a path-finding algorithm, based on structured data such as node-to-node distances and path connectivity collected during fieldwork.

AR navigation cues, including markers, are rendered using Unity and ARCore, ensuring realistic guidance for outdoor navigation.

The application also includes navigation audio support, which provides audio prompts that announce the name of a building or infrastructure when the user reaches intermediate points or the destination along the route. These audio prompts are intended to assist users during navigation and enhance accessibility, particularly by confirming location progress without requiring constant visual attention.

To ensure the system remains accurate and up to date, a web-based admin panel is included. Authorized WMSU staff can use this interface to manage buildings, walkable paths, distances, and other campus elements. All map data is stored in a node-based graph structure, and updates are synced via a Firebase database to ensure the app always reflects the current campus layout.

Limitation

- **Navigation Coverage:** The system currently supports WMSU Campus A and Campus B, which are physically connected. Campus C is in a separate location; although users can select it as a navigation area, navigation between Campus A/B and Campus C is not available due to the geographical separation.
- **GPS Dependence:** Navigation accuracy relies heavily on GPS signal strength. GPS may become unstable or less accurate in crowded, covered, or obstructed areas, potentially causing AR positioning to drift.
- **QR-Based Recalibration:** QR codes are used only to recalibrate the user's position when GPS signals are weak or inaccurate. Real-time navigation is not provided via QR codes. QR codes must be physically accessible, properly placed, undamaged, and scannable for the system to restore accurate location data.
- **AR and Device Limitations:** Visual stability and AR performance may vary depending on the device model. Lower-end devices may experience reduced performance, and sensor accuracy (e.g., compass and gyroscope) may differ across phones.
- **Navigation Audio Limitations:** Audio prompts are limited to announcing the name of the building or infrastructure when the user reaches intermediate nodes or the destination. The system does not provide turn-by-turn voice instructions, does not indicate left or right directions, and does not announce nearby buildings or landmarks that are not part of the selected route.
- **Indoor Navigation:** Indoor floor plans are only accessible via QR code scans. Navigation guidance within buildings is visual only, as the app does not provide

directional tracking, audio, or voice instructions. An Indoor button appears when a scanned building has a floor plan, allowing users to switch the map view.

- Platform Requirements: This application is Android-only and relies on ARCore for AR functionality.

The device must support ARCore (official list: <https://developers.google.com/ar/devices>).

The device must run Android 7.0 (API level 24) or higher to access full AR features.

Significance of the Study

This project provides a modern, technology-driven solution to the challenge of campus navigation. By utilizing AR and real-time route guidance, the system enhances the overall accessibility, efficiency, and orientation of users across campus. The following stakeholders are expected to benefit from this system:

- Freshmen and New Students

Will find it easier to locate buildings, classrooms, and offices during their first weeks in the university, reducing anxiety, confusion, and late arrivals. The QR code functionality also allows new students to identify their starting point even without GPS.

- Visitors and Guests

Will be able to confidently navigate the campus without needing constant assistance or asking for directions, especially during university events, enrollment,

or public seminars. Scanning QR codes placed around the campus provides immediate location recognition without requiring an internet connection.

- University staff and Administrators

Will benefit from reduced time spent assisting lost visitors and can maintain and update the campus route network easily through the admin interface.

- Event attendees and Applicants

Will be guided to specific venues (e.g., Gym, Registrar, Auditorium) more efficiently during campus-wide activities, examinations, or job fairs.

- The university as an institution (WMSU)

Gains a modern and tech-driven campus support system that reflects innovation, improves visitor experience, and supports its digital transformation efforts.

- Capstone and IT Students

Gain practical experience in real-world AR development, GPS tracking, pathfinding, and user-centered design, all applicable in smart city and navigation technology.

Definition of Terms

Table 1: Definition of Terms

Term	Definition
1. Augmented Reality (AR)	A technology that overlays digital content such as images, arrows, or labels onto the real-world view through a smartphone or AR device. In this project, it provides navigation cues in real time. (Source: [21] Oxford University Press, 2023)
2. GPS (Global Positioning System)	A satellite-based navigation system that provides location and time information to a device anywhere on Earth. It enables the app to detect the user's real-time location outdoors. (Source: [22] U.S. Department of Defense GPS.gov, 2023)
3. A-GPS (Assisted GPS)	A system that uses internet or mobile data to speed up the GPS signal acquisition process, improving location accuracy and reducing delay in determining initial location. (Source: [23] Android Developers, 2024)
4. Pathfinding Algorithm	A computational method is used to find the most efficient path between two points. In this project, it helps determine the shortest route between buildings. (Source: [24] Russell & Norvig, 2021)
5. Dijkstra's Algorithm	A well-known path-finding algorithm that calculates the shortest path between nodes in a graph. It is used in the app to determine the most efficient route to the selected destination. (Source: [25] Dijkstra, 1959)
6. Node	A point in a network or graph representing a location, such as a building or checkpoint in the WMSU campus. (Source: [26] Techopedia, 2024)
7. Edge	A connection between two nodes representing a walkable path between two buildings, along with the distance in meters. (Source: [27] Techopedia, 2024)
8. Firebase	A cloud-based platform by Google that provides real-time database, storage, and syncing services. It is used to

Term	Definition
	store and update the campus map data in this project. (Source: [28] <i>Firebase Documentation</i> , 2024)
9. Unity	A cross-platform game engine used to develop 2D and 3D applications, including AR apps. It is the primary development tool for building the mobile navigation system. (Source: [29] <i>Unity Technologies</i> , 2024)
10. AR Foundation	A Unity framework that supports the development of cross-platform AR experiences by combining ARKit (iOS) and ARCore (Android) features. (Source: [30] <i>Unity Docs</i> , 2024)
11. Real-Time Tracking	The continuous monitoring of a user's position during movement using GPS, allowing the app to update the navigation path dynamically. (Source: [Proponent's Definition], 2024)
12. Infrastructure	Refers to the underlying campus structures, buildings, pathways, and physical layout represented within the system. It includes the digital mapping of roads, landmarks, building entries, and floor plans that enable the application to generate routes and support navigation. (Source: <i>Proponents' Definition</i> , 2024; supported by Haq et al., 2017)
13. Digital Wayfinding	The process of navigating through a physical environment using digital tools such as maps, arrows, or route instructions provided via mobile applications. (Source: [31] <i>Smart Cities Journal</i> , 2023)
14. Overpass	A raised structure that allows users to cross from Campus A to Campus B without interacting with ground-level traffic or obstructions. (Source: <i>Proponent's Definition</i> , 2024)
15. Navigation Mode	Referring to the method the app uses to determine and start a route. In this project, it includes GPS-based (online) and manual selection (offline). (Source: <i>Proponent's Definition</i> , 2024)

CHAPTER II

REVIEW OF RELATED LITERATURE

Project Context

Reliable and user-friendly navigation in university campuses has become essential due to the complexity of modern educational facilities. Students, staff, and visitors at Western Mindanao State University (WMSU) in Zamboanga City face difficulties locating classrooms, offices, and facilities across multiple buildings and outdoor areas. Traditional GPS-based maps are often inaccurate indoors, while AR-based systems provide real-time, interactive guidance that improves orientation and wayfinding [1], [2].

Existing solutions have limitations. Many depend on internet connectivity, BLE beacons, or Wi-Fi fingerprinting, which are less effective in low-signal areas [5], [6], [11]. GPS-based methods fail indoors, and current AR systems often lack seamless transitions between indoor and outdoor guidance. CRIMSON MAP addresses this gap with a hybrid AR navigation system that integrates GPS-based outdoor mapping and QR code-based indoor localization, ensuring accurate overlays even under low lighting, dynamic crowds, or campus infrastructure changes [1], [6], [7].

Recent literature informs CRIMSON MAP's design. ARCore-based 3D maps with visual-inertial odometry support indoor-outdoor navigation [1]; deep learning enhances AR target recognition [2]; accessible systems for visually impaired users improve usability [3]; SLAM-based methods allow indoor localization without infrastructure [5], [6]; and Semantic Web frameworks provide context-aware guidance [11]. Drawing from these studies, CRIMSON MAP combines hybrid mapping, robust localization, and QR code-based indoor positioning to deliver a comprehensive navigation solution.

The expected outcome is a fully functional AR-based navigation system that seamlessly guides users across WMSU campuses. By integrating GPS-based outdoor routing, indoor QR localization, and user-centered design principles, CRIMSON MAP enhances wayfinding, accessibility, and overall campus navigation for students, staff, and visitors [1], [7], [11].

Related Studies

The Review of Related Literature provides the foundation for CRIMSON MAP, an AR navigation system designed to simplify wayfinding across WMSU's Main Campus, including Campuses A, B, and C. Built using Unity and ARCore, CRIMSON MAP delivers real-time, intuitive directions with GPS-based outdoor navigation for reliable guidance even in areas with limited connectivity.

Key studies examined include AR navigation systems, localization techniques, user experience evaluation, and 3D model placement methods. These insights inform CRIMSON MAP's features, including hybrid indoor-outdoor mapping, QR code-based indoor positioning, robust real-time overlays, and accessibility considerations, ensuring the system is accurate, efficient, and user-friendly for the WMSU community.

2.1.1 ARCore-Based Augmented Reality Campus Navigation System



Figure 1: 3D Campus Map from the ARCore-Based Augmented Reality Campus Navigation System (Lu et al.,2021)

Figure 1 shows a 3D campus map generated by the ARCore-based Augmented Reality Campus Navigation System developed by Lu et al. (2021), presenting a detailed virtual representation of a university environment integrated with augmented reality navigation features. The image illustrates how digital 3D guidance can be overlaid onto real-world surroundings to assist users in navigating complex campus layouts.

Lu et al. [1] introduced an ARCore-based campus navigation system that overlays 3D directional guidance onto real-world scenes using visual-inertial odometry to improve positioning accuracy. Their system demonstrated effective outdoor navigation performance; however, it relied heavily on stable lighting conditions, precise sensor calibration, and consistent environmental features for reliable AR tracking. A key limitation relevant to CRIMSON MAP is that the system requires stable internet connectivity and lacks an offline fallback mechanism, making it less suitable for campuses with weak GPS reception.

or limited internet, conditions addressed by the CRIMSON MAP system.

2.1.2 SLAM-based Indoor Navigation in University Building



Figure 2: The final version of the AR navigation application

As shown in Figure 2, the augmented reality (AR) navigation application developed by Chidsin et al. (2021) presents a real-time camera view enhanced with overlaid room labels and directional arrows that guide users through indoor environments. The system, designed for use with smart glasses, demonstrates how AR elements can provide intuitive and context-aware navigation assistance within complex indoor spaces.

Sukhareva et al. [6] developed a SLAM-based indoor navigation system that overlays AR directions onto live camera views, enabling room-to-room guidance without relying on GPS or external hardware. While the approach achieved high accuracy in stable and well-lit environments, its performance decreased in dim areas, visually uniform hallways, or crowded indoor spaces where tracking becomes challenging. These limitations emphasize the relevance of CRIMSON MAP's design choice to integrate QR-assisted floor plan access with GPS-first navigation, allowing more consistent performance even when SLAM conditions are not ideal.

2.1.3 Sensor-based Indoor Navigation with Augmented Reality and Semantic Web

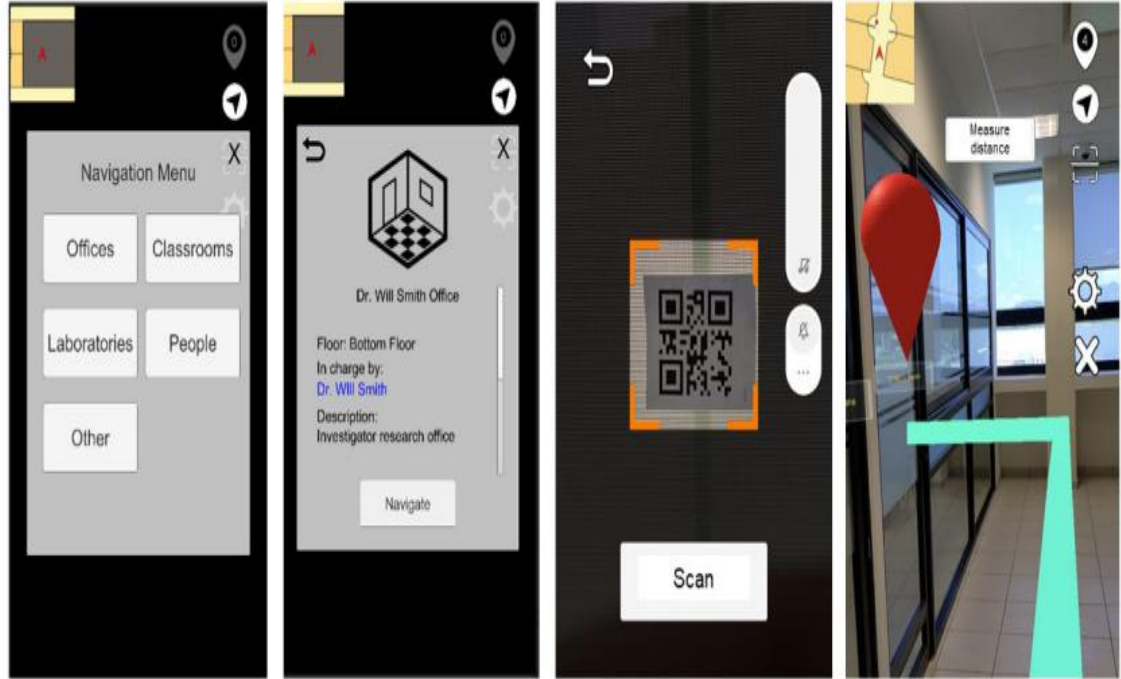


Figure 3: Developed Application's Interfaces Examples

As shown in Figure 3, the developed application demonstrates interface examples that integrate augmented reality overlays with navigation features designed to assist users within campus environments. The interfaces highlight how visual guidance elements can improve user interaction and spatial orientation during navigation tasks.

Rubio-Sandoval et al. [11] developed an indoor navigation system that combines AR overlays with a Semantic Web knowledge base to guide users through university buildings without relying on continuous GPS connectivity. Although effective, the system depends on QR codes for initial positioning and recalibration and experiences tracking errors in nearly 30% of navigation tasks due to lighting conditions and feature detection limitations. These constraints reinforce CRIMSON MAP's design approach of GPS-first outdoor navigation complemented by QR-assisted indoor recalibration, enabling more stable and accessible guidance despite varying network and environmental conditions.

2.1.4 AR-Based Navigation Using RGB-D Camera and Hybrid Map



Figure 4: Navigation with Direction Arrow

Figure 4 illustrates a navigation interface featuring a detailed 3D model of a college department's interior, including hallways, escalators, and other key structural elements that support spatial awareness. Realistic textures and lighting enhance environmental visualization, while clearly defined paths indicate possible navigation routes within the space. The inclusion of an "Area Target Behavior" script further suggests integration with augmented reality interaction, enabling dynamic and interactive viewing during navigation.

Chidsin et al. [6] developed an AR navigation system utilizing SLAM technology with an RGB-D camera and a hybrid mapping approach, achieving high accuracy in directional guidance and feature labeling for indoor environments. However, the system relies on relatively static surroundings and is limited to single-building implementation, reducing adaptability in larger or more dynamic campus settings. CRIMSON MAP expands this concept by extending navigation across the entire WMSU campus,

integrating GPS-based outdoor guidance with indoor recalibration to maintain reliable navigation even in high-traffic and multi-building environments.

2.1.5 A Survey on Improving Navigation for Street Data Using Mobile Augmented Reality

Early mobile AR navigation systems displayed points-of-interest (POI) but required users to switch between AR and 2D maps, reducing route continuity and spatial awareness [1]. Recent studies integrate sensor fusion, landmark recognition, and hybrid localization to enhance real-time orientation and overlay accuracy [2], though scalability, energy efficiency, and adaptability remain challenges, particularly in complex or dynamic environments [7]. CRIMSON MAP addresses these gaps by providing continuous AR route visualization, campus-specific guidance, QR-based initialization for recalibration, and accessibility features tailored to students, staff, and visitors, eliminating the need to alternate between AR and traditional maps [1].

2.1.6 Issues and Challenges in Handheld Augmented Reality Applications

Early AR navigation relied on 2D maps and textual cues, forcing users to switch views and hindering seamless route comprehension [8]. Advances now integrate live camera feeds with GPS, accelerometers, and compasses to provide real-time, context-aware guidance, yet handheld AR still faces challenges in tracking accuracy, hardware limitations, and usability under varying environmental conditions [8], [9]. These gaps are particularly evident indoors, where GPS signals are weak, highlighting the need for campus-specific solutions. CRIMSON MAP addresses these issues by combining AR overlays with QR code markers and node-based mapping, enabling reliable navigation for both indoor and outdoor WMSU campus routes.

2.1.7 A Review of VSLAM Technology Applied in Augmented Reality

Marker-based tracking in early AR systems limited accuracy and real-time performance, but visual simultaneous localization and mapping (VSLAM) now allow simultaneous environment mapping and user localization, improving robustness and precision [4]. Hybrid VSLAM frameworks, combining visual odometry, loop closure, and optimization, reduce cumulative errors and enhance global consistency, while methods like ORB-SLAM and LSD-SLAM balance computational cost and accuracy [4]. Despite these advances, challenges persist in low-texture, repetitive, or poorly lit environments, and handheld systems still face usability issues under varying conditions [4], [9]. CRIMSON MAP addresses these limitations by integrating VSLAM with QR code anchors and node-based maps, ensuring consistent indoor and outdoor navigation across the WMSU campus.

2.1.8 User Experience Evaluation of Augmented Reality: Applications in Navigation

Mobile AR navigation research emphasizes seamless integration of digital overlays with real-world environments, where usability, intuitiveness, and cognitive load strongly influence user performance and acceptance [7]. User experience (UX) evaluation systematically assesses aspects like ease of use, efficiency, and satisfaction, yet prior studies reveal inconsistencies in evaluation methods, limited long-term engagement analysis, and a lack of standardized protocols [7], [4]. These limitations are particularly relevant for CRIMSON MAP, which addresses campus-specific navigation at WMSU by incorporating standardized usability metrics, context-aware evaluations, and interface designs that ensure consistent user experience across diverse indoor and outdoor environments.

2.1.9 Augmented Reality Navigation with Image Segmentation and Sensor Fusion

Integrating image segmentation with sensor fusion has enhanced AR navigation precision by combining visual data with GPS, gyroscope, and accelerometer inputs [8]. Zhang et al. [8] demonstrated that GA-OTSU-Canny segmentation with multi-sensor fusion improves building recognition and real-time tracking, while Graser et al. [7] highlighted the need to balance technical accuracy with usability. Limitations include reduced reliability in dense campus layouts and connectivity-constrained environments, challenges directly addressed by CRIMSON MAP through efficient sensor fusion, image processing, and user-centered interface design tailored for WMSU.

2.1.10 Geolocation-Based Augmented Reality with UnityGeoAR

Geolocation AR frameworks like UnityGeoAR improve AR navigation by integrating GPS data with 3D visualization, handling coordinate reprojection, camera alignment, and elevation mapping within Unity3D [10]. While effective in anchoring digital elements to the user's environment, these systems often lack standardization, rely on isolated solutions, and exhibit sensor drift, limiting accuracy in campus-like settings [10]. CRIMSON MAP leverages this approach for precise spatial alignment while incorporating systematic user experience evaluation to ensure usability and reliability for WMSU students and staff [7], [10].

2.1.11 Augmented Reality Navigation

AR navigation enhances traditional maps by overlaying digital cues onto real-world views, using smartphones, GPS, sensors, and cameras to provide contextual guidance for indoor and outdoor environments [6]. While these systems improve pathfinding and

reduce screen distraction, challenges remain in device compatibility, privacy, and potential visual overload from excessive information [6]. CRIMSON MAP builds on this approach by integrating real-time geolocation with intuitive AR overlays, specifically tailored for WMSU campus navigation to improve accessibility and user experience [6].

2.1.12 Research-Efficient AR Navigation Using Crowdsourced Data

Crowdsourced AR navigation reduces reliance on large spatial databases by using lightweight data formats and simplified storage, enabling accurate route guidance with lower CPU and memory usage [8]. Singh et al. [8] demonstrated up to 99% localization accuracy while reducing processing demands, employing algorithms like A* and nearest-neighbor search for efficient real-time navigation. Limitations include data quality maintenance and timely updates in dynamic environments, which CRIMSON MAP addresses by combining lightweight spatial structures with campus-specific indoor and outdoor mapping to ensure smooth performance on mobile devices [8].

2.1.13 Assessment of 3D Models Placement Methods in Augmented Reality

Accurate placement of 3D models in AR remains a major challenge due to sensor inaccuracies, camera instability, and misalignment between real and virtual environments [9]. El Barhoumi et al. [9] evaluated marker-based, marker-less, sensor-based, and hybrid methods, reporting placement errors from centimeters to meters depending on the technique, with marker-based approaches offering high precision but limited flexibility, and marker-less methods being more adaptable but less consistent. These limitations are relevant to CRIMSON MAP, which requires precise placement of navigation cues and emergency markers across WMSU's campus; hybrid or vision-based approaches can enhance stability and reliability for both indoor and outdoor navigation [9].

2.1.14 University Campus Indoor Navigation Using AR and A* Search Algorithm

Alsabbagh et al. [11] developed an indoor navigation system combining augmented reality with the A* search algorithm to guide Android users in GPS-denied university buildings. The system features virtual path indicators, customizable location menus, and image-based initialization, reducing reliance on infrastructure such as BLE beacons or Wi-Fi fingerprinting. Limitations included occasional misalignment with walls, highlighting the need for precise 3D modeling, which CRIMSON MAP addresses by integrating accurate AR overlays and campus-specific indoor mapping for seamless navigation [11].

2.1.15 Analysis of Dijkstra's and Bellman-Ford Algorithm in Passenger Flow and Tweet Changes

Shortest path algorithms like Dijkstra's and Bellman-Ford have been widely used to optimize travel in complex networks, with Dijkstra providing faster computation and Bellman-Ford offering flexibility for graphs with negative edges [12]. Prior studies focused on metro systems using smart card and social media data to analyze passenger flow, leaving limited research on campus-scale applications. For CRIMSON MAP, adapting Dijkstra's algorithm enhances efficient route guidance across WMSU pathways, especially during emergencies, addressing the gap in campus-specific navigation [12].

2.1.16 Integrating User Stories in the design of Augmented Reality Application

In AR development, user stories and related tools like personas and journey maps enable developers to capture user needs effectively, improving requirement elicitation and system alignment with expectations [13]. Ankora and Aju applied iterative Design Science Research, refining AR applications through feedback to enhance usability and relevance. For CRIMSON MAP, integrating user stories ensures the system addresses campus-

specific navigation scenarios, such as locating classrooms or emergency exits; however, existing studies show gaps in sustaining long-term engagement and combining qualitative insights with quantitative performance metrics, which CRIMSON MAP must consider [13].

2.1.17 Improved A* Navigation Path-Planning Algorithm Based on Hexagonal Grid

Traditional square grid-based path-planning often produced rigid routes with high computational costs, prompting advancements such as the hexagonal grid-based A* algorithm by An, Rui, and Gao [14]. This method improves trajectory smoothness, reduces unnecessary turns, and incorporates adaptive heuristics and turning penalties for more efficient path generation. However, its validation remains limited to simulations, highlighting the need for real-world testing; for CRIMSON MAP, integrating such techniques can enhance campus navigation by generating smoother, safer, and more accurate routes, particularly in irregular or obstacle-rich areas [14].

2.1.18 Augmented Reality and GPS-Based Resource Efficient Navigation System for Outdoor Environments: Integrating Device Camera, Sensors, and Storage

Traditional AR navigation often relied on large proprietary maps, consuming significant storage and processing resources, limiting scalability. Singh et al. [15] introduced a resource-efficient system integrating device cameras, sensors, and lightweight crowdsourced spatial data, achieving high accuracy while reducing memory and CPU load. Limitations include reliance on outdoor GPS signals and potential inconsistencies in crowdsourced data; for CRIMSON MAP, adopting these lightweight strategies supports accurate indoor and outdoor campus navigation without overloading devices, while also enhancing offline usability and accessibility in low-connectivity areas [15].

2.1.19 A Study on Bellman Ford Algorithm for Shortest Path Detection in Global Positioning System

Shortest path algorithms are crucial for navigation, with Dijkstra excelling in speed and Bellman-Ford handling negative edge weights for complex networks. Rai [16] applied Bellman-Ford to GPS navigation, showing it can find more optimal routes than Dijkstra in certain scenarios, but with higher computational costs. For CRIMSON MAP, leveraging Bellman-Ford could improve campus route optimization across indoor and outdoor paths; however, balancing accuracy with processing efficiency will require hybrid strategies or algorithmic refinements to ensure real-time performance [16].

2.1.20 Adaptive AR Navigation: Real-Time Mapping for Indoor Environment Using Node Placement and Marker Localization

Indoor AR navigation increasingly uses node placement and marker localization to create real-time maps, enabling dynamic pathfinding with ARCore, NavMesh, and A* without pre-built models [17]. While effective for adaptability and user interaction, limitations include interface clarity issues, physical strain during use, and scalability challenges in larger spaces. For CRIMSON MAP, adopting node-based AR mapping can enhance flexible, campus-wide indoor navigation, but careful attention is needed to maintain usability and efficiency across diverse buildings while complementing algorithmic route planning like Bellman-Ford [16], [17].

2.1.21 Outdoor Navigation System by AR

Outdoor AR navigation integrates real-time visual guidance directly into the user's environment, using tools like Unity, AR Foundation, and ARCore to overlay arrows, labels, and distances on camera feeds [20]. While this approach improves spatial awareness

compared to conventional GPS maps, limitations include GPS inaccuracies, variable environmental conditions, and the need to optimize device performance for smooth rendering [20], [15]. For CRIMSON MAP, these insights inform the integration of GPS-based outdoor guidance with indoor node and marker-based mapping, ensuring seamless campus-wide navigation while maintaining usability and efficiency across mobile platforms [16], [17], [13].

Table 2: A comparison table between Dijkstra, Bellman-Ford, and A* Algorithm

Criteria	Dijkstra	Bellman-Ford	A* (A-Star)
Uses Heuristics (Estimates)	No	No	Yes
Speed for Large Maps	Fast	Slow	Fastest (with accurate heuristics)
Supports Negative Weights	No	Yes	No
Path Accuracy	Optimal	Optimal	Optimal (if heuristic is good)
Best Use Case	Simple graphs without estimates	Graphs with possible negative edges	Navigation with estimated distances (like maps)
Suitable for Real-time Navigation	Yes	No (Too slow)	Yes (Best option)

The table compares three path-finding algorithms: Dijkstra, Bellman-Ford, and A*, based on how they perform in different situations. One key difference is that A* uses something called a heuristic, which is like an estimate or a "smart guess" of how far a point is from the destination. This allows A* to focus on the most promising paths first, making it faster than the other two, especially in large maps like a university campus. Dijkstra also finds the shortest path, but it checks all possible directions without using any estimate, which can take more time. Bellman-Ford can handle negative path values (which means it can deal with certain mathematical edge cases), but it's much slower and not practical for real-time use like navigation apps. A* is also accurate, it still finds the shortest path, just more efficiently because it uses both the actual distance and the estimated distance to the goal. That's why for an app like CRIMSON MAP, which helps students navigate around campus quickly, A* is the best choice. It's fast, accurate, and perfect for map-based systems where the user's current location and destination are known.

Synthesis

Table 3: Synthesis

Features	ARCore-Based AR-CNS (Lu et al., 2021)	SLAM-based Indoor Navigation (Sukhareva et al., 2021)	Sensor-Based AR (Rubio- Sandoval et al., 2021)	AR-Based RGB-D (Chidsin et al., 2021)	CRIMSON MAP(Proposed)
AR Navigation Cues	✓	✓	✓	✓	✓
Shortest Path Calculation		✓			✓
QR Code Positioning		✓	✓		✓
Offline Navigation Mode		✓		✓	✓
Admin Map Updates via Database	✓		✓		✓

Node-Based Campus Map	✓	✓	✓		✓
Indoor Floor Plan					✓
Recalibration via QR Codes					✓

Table 3: it is evident that no single existing system provides a comprehensive solution across all dimensions. For instance, ARCore-based navigation systems (Lu et al., 2021) emphasize AR cues and map updates but lack shortest path algorithms and QR-based localization. SLAM-based indoor navigation (Sukhareva et al., 2021) excels in offline mode and QR code positioning but is computationally intensive. Sensor-based AR approaches (Rubio-Sandoval et al., 2021) focus on integrating device sensors with QR support, though their adaptability across campus settings is limited. AR-based RGB-D methods (Chidsin et al., 2021) offer robust indoor positioning with offline capabilities but require specialized hardware.

In contrast, CRIMSON MAP combines the strengths of these approaches by offering AR navigation cues, shortest path calculation, QR code positioning, offline functionality, admin-managed database updates, a node-based campus map, indoor floor plans, and recalibration via QR codes. This synthesis highlights how CRIMSON MAP advances AR navigation research by integrating essential features into a single, adaptable system specifically designed for university campus navigation.

Conceptual Framework

The conceptual framework of CRIMSON MAP, the AR Campus Navigation System, demonstrates the structure and workflow of a mobile application designed to enhance indoor and outdoor wayfinding across the Western Mindanao State University (WMSU) Main Campus. The system provides students, staff, and visitors with real-time AR-based guidance between buildings and facilities, integrating GPS, QR codes, and sensor data for accurate positioning.

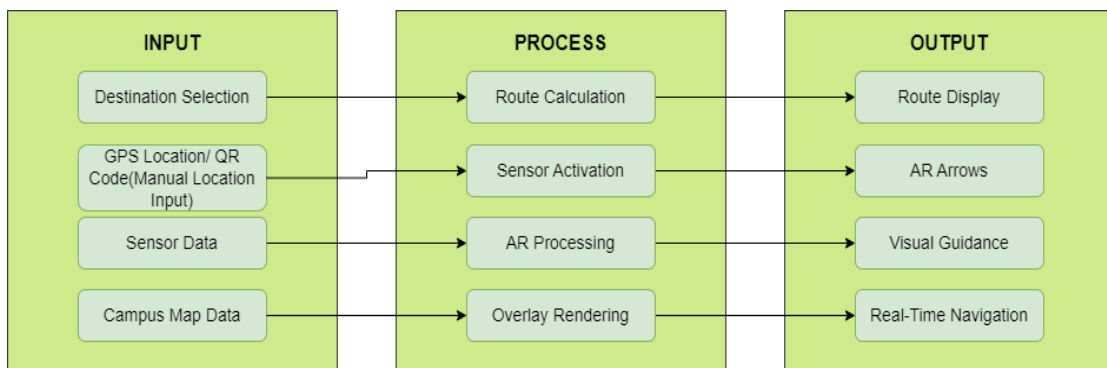


Figure 5: Conceptual Framework

Grounded in the Input–Process–Output (IPO) model, the framework illustrates how user selections, location data, and device sensor inputs are processed to generate dynamic visual navigation overlays. The Input phase includes user-selected destinations, GPS coordinates, QR code scans for indoor locations, and sensor data from accelerometers, gyroscopes, and cameras. Pre-stored campus map data, represented as nodes (locations) and edges (walkable paths), provides the spatial reference for route computation.

The Process phase encompasses route calculation, tracking, and AR rendering. The system computes optimal paths using the A* algorithm for outdoor navigation and combines hybrid localization techniques such as visual-inertial odometry (VIO) and QR code positioning for indoor guidance. Sensor data continuously updates the user's position

and orientation, while Unity and ARCore render interactive AR overlays that display directional cues, waypoints, and distance markers. These operations occur sequentially and in parallel to maintain responsive and context-aware navigation across dynamic campus environments.

The Output phase delivers an immersive navigation experience. Visual route recommendations, AR markers, and dynamic overlays guide users step by step, adapting to real-time movement and environmental changes. The system highlights optimal paths while ensuring seamless indoor-outdoor transitions. This approach improves accessibility, reduces navigation-related confusion, and supports a user-friendly experience for freshmen, visitors, and staff navigating WMSU's extensive campus.

CHAPTER III

METHODOLOGY

Research Design

This study adopts a Design and Development Research (DDR) approach, guided by the Agile development methodology. DDR offers a structured framework for designing, refining, and evaluating technology-based solutions, with an emphasis on iterative improvement and regular feedback cycles. This approach ensures that the system meets both technical and user-centric requirements effectively.

The research process begins with a problem analysis, which identifies the limitations of traditional navigation tools (such as Google Maps) in indoor and campus-specific settings. These limitations highlight the need for a customized navigation solution that supports GPS-based, with augmented reality (AR) overlays to enhance the user experience and provide real-time guidance.

Guided by Agile principles, the system is developed in iterative sprints. Each sprint results in a working component of the system, which undergoes usability testing and is refined based on stakeholder feedback. This iterative process ensures that the system evolves to meet user needs and addresses emerging technical requirements.

Data Source

This study will utilize both primary and secondary data sources to ensure the accuracy, reliability, and relevance of the information needed in the development of the CRIMSON MAP AR navigation system for Western Mindanao State University (WMSU).

Primary data will be obtained through on-campus observations and technical measurements. Visual and observational data will be gathered on-site to validate the accuracy of building locations, walkways, and QR code placements across the campus. Technical data will include the geographic coordinates (longitude and latitude) of buildings and landmarks, which will be collected using the GPS Test application and verified with Google Maps. Physical distances between nodes, as well as distances of rooms within buildings, will be measured manually and further validated with Google Earth. These coordinates and measurements will then be converted into Unity's Cartesian coordinate system for use in implementing AR navigation and pathfinding algorithms.

Secondary data will come from existing campus layout references such as official WMSU maps and publicly available digital maps. Related literature and previous studies on augmented reality navigation, QR-based positioning, and campus-specific systems will also be reviewed to guide the design and validation of CRIMSON MAP. Furthermore, technical documentation from Unity and Firebase, as well as online resources such as GPS tools and coordinate converters, will serve as supplementary sources of information.

To ensure accuracy, technical measurements will be cross validated using multiple tools, including GPS Test, Google Maps, and Google Earth, alongside manual observations. This combination of primary and secondary sources will provide a comprehensive and reliable dataset for the development and evaluation of the CRIMSON MAP system.

Data Gathering Instrument

The CRIMSON MAP system relies on a combination of technical tools and applications to ensure accurate, reliable, and validated location-based information for both

indoor and outdoor navigation. Each tool was selected based on its suitability, precision, and compatibility with AR navigation development:

- **GPS Test Application:** This tool is used to obtain the geographic coordinates (latitude and longitude) of buildings, pathways, and key landmarks within the WMSU campus. It was chosen for its ability to provide real-time, on-site location data, which forms the foundation of GPS mode.
- **Google Maps:** Google Maps serves as a reference platform to verify the accuracy of the collected coordinates and cross-check the campus layout. Its high-resolution mapping and satellite imagery ensure that the AR system is aligned with real-world locations.
- **Google Earth:** Google Earth is employed to validate distances between buildings and campus areas requiring confirmation. It provides precise geospatial measurements, ensuring that the coordinates and paths in the system reflect actual campus distances.
- **Fiberglass Measuring Tape (100 meters):** Manual measurement using a fiberglass tape allows for precise calculation of distances between buildings and indoor rooms. This is essential for establishing accurate x and y coordinates for offline navigation, especially in areas where GPS signals are weak or unavailable.
- **Unity (Game Engine):** Unity is used to translate the gathered geographic coordinates into a Cartesian coordinate system compatible with the AR navigation system. Its robust AR development tools, including integration with ARCore, make it ideal for building interactive and immersive AR navigation experiences.
- **Firestore (Firebase):** Firestore provides a cloud-based database for storing and retrieving campus data, including building details, room information, and coordinates. Its real-time syncing capabilities ensure that updates made by

administrators are immediately available to the mobile app, supporting dynamic and accurate navigation.

- **Software Evaluation Form:** This instrument is used to collect user feedback during testing. It evaluates usability, accuracy, clarity, and overall satisfaction with the system. This ensures that the AR navigation application meets user needs and provides a reliable, user-friendly experience.

Together, these tools and instruments provide a comprehensive, validated, and reliable dataset that enables the CRIMSON MAP system to deliver accurate AR-based navigation across both indoor and outdoor areas of the WMSU campus.

Data Gathering Technique and Procedures

Techniques

To gather the needed data for the development and validation of CRIMSON MAP, the following techniques were applied:

1. **GPS Data Collection** – Used GPS Test application to obtain geographic coordinates (longitude and latitude) of buildings, rooms, and landmarks inside the WMSU campus.
2. **Digital Mapping** – Verified and cross-checked coordinates using Google Maps and Google Earth to ensure accuracy of building locations and pathways.
3. **On-Site Observation and Measurement** – Conducted physical walkthroughs inside the campus to identify landmarks, measure approximate distances (with manual tools when needed), and document building layouts.
4. **Document Review** – Referred to available WMSU campus maps and building references to support accuracy of location data.

5. System Integration and Testing – Converted gathered coordinates into Unity’s Cartesian coordinate system and tested navigation functions using Firestore (Firebase) as the database for storing and retrieving building and route information.
6. User Testing – Conducted usability testing involving freshmen participants to evaluate the accuracy, responsiveness, and user experience of the system.

Procedures

The data gathering will follow these steps:

Step 1: Preliminary Preparation – The researchers will conduct initial walkthroughs of the WMSU Main Campus to identify target buildings, pathways, and landmarks to be included in the navigation system.

Step 2: Coordinate Collection – The GPS Test application will be used to gather the latitude and longitude of key locations such as buildings, entrances, rooms, and important nodes within the campus.

Step 3: Data Verification – The collected GPS data will be cross-checked with Google Maps and Google Earth to confirm the accuracy of location points, layout, and distances.

Step 4: Distance Measurement – A 100-meter fiberglass measuring tape will be used to manually measure distances between nodes (e.g., rooms, hallways, and building entrances) in meters. These measurements will be used to establish accurate x and y coordinates to improve pathfinding precision.

Step 5: Data Encoding and Storage – All collected data will be organized and encoded into Firestore (Firebase), which will serve as the central database for building details, coordinates, and navigation pathways.

Step 6: System Integration and Testing – The verified coordinates will be implemented in Unity by converting them into Cartesian coordinates. The researchers will then test the AR

navigation system to validate route accuracy, building placement, and system responsiveness in GPS Mode

Step 7: Human Participant Evaluation – 30 freshmen students from various departments of WMSU participated in usability testing. Each participant will be asked to use the system to navigate between selected buildings and complete a software evaluation form measuring ease of use, accuracy, and satisfaction.

Before conducting user testing, participants are asked to sign an informed consent form approved by the researchers and the adviser. This would ensure that participation was voluntary and that all responses would remain confidential and used solely for research and evaluation purposes.

Step 8: Documentation and Analysis – All gathered and validated data were compiled, reviewed for consistency, and analyzed for accuracy and usability following the DDR and Agile methodology framework.

Data Analysis

The data to be collected for CRIMSON MAP will be systematically organized and analyzed to ensure accuracy, consistency, and usability in the development of the AR navigation system. The analysis will be conducted in the following way:

1. GPS Data Analysis – The geographic coordinates (latitude and longitude) gathered from the GPS Test application will be checked, cleaned, and compared with Google Maps and Google Earth to ensure accuracy and reliability.

2. Distance and Route Analysis – The measured and estimated distances between nodes (e.g., rooms, buildings, and entrances) will be verified and converted into Unity's Cartesian coordinate system for use in the pathfinding algorithm.
3. System Testing Data – The integrated navigation system will be tested in Unity using Firestore (Firebase) data. The accuracy of building placements, navigation routes, and distances will be validated against the actual WMSU campus layout.
4. Validation – The collected data will be cross-checked with official WMSU campus maps and on-site observations to confirm correctness. Consistency across different tools (GPS Test, Google Maps, Google Earth, and manual checks) will be ensured before final integration.

Software Development

CRIMSON MAP will be developed as a mobile Augmented Reality (AR) navigation system using the Agile development methodology, ensuring iterative design, testing, and refinement across multiple sprints. The system will utilize Unity as the main development platform, Google ARCore for AR capabilities, and Firebase for real-time cloud database management and synchronization.

The development process will include the following key components and integrations:

1. AR Framework Integration – Google ARCore will be integrated with Unity to handle motion tracking, environmental understanding, and light estimation. These features will enable the system to place and anchor virtual objects in real-world coordinates with high precision.
2. Database and Backend Integration – Firebase (Firestore) will serve as the centralized database to store all coordinates, building details, and pathway data.

JSON-based data synchronization will ensure that any updates made via the web-based admin dashboard are automatically reflected in the mobile app.

3. Location and Marker Detection – The AR navigation system will primarily rely on GPS-based location tracking to position virtual markers. A fallback QR code scanning feature will also be developed to allow manual location detection in areas where GPS accuracy is low.
4. 3D Visualization and Optimization – Unity’s built-in 3D primitives (such as cubes and planes) will be customized to represent buildings and markers. Lightweight textures and shaders will be used to maintain smooth performance even on lower-end Android devices. Techniques such as occlusion culling and static batching will be applied to further optimize rendering.
5. Lighting, Scaling, and Anchoring – ARCore’s light estimation feature will automatically adjust the brightness of virtual elements to match real-world lighting. Anchors will be used to fix virtual objects to real-world positions, ensuring realistic rendering and stability during navigation.
6. Cross-Platform Scalability – The initial release will support Android devices compatible with ARCore. Future updates will plan for iOS integration via ARKit to expand device compatibility.
7. System Logic and Synchronization – The app will fetch and locally cache map data from Firebase in JSON format. Upon startup, the system will check for updates and automatically synchronize to ensure the latest map and pathfinding data are available both online and offline.
8. User Interface and Navigation Flow – A simple and interactive interface will be developed within Unity, featuring intuitive buttons, AR markers, and smooth transitions between map and AR modes.

Process Flow Overview:

1. Location Selection – Users will select their target building or facility within the campus.
2. Coordinate Mapping – The system will retrieve stored coordinates and compute the best route using the A* pathfinding algorithm.
3. AR Navigation – The app will overlay directional markers and guidance paths in the AR view to assist the user in real-time.
4. System Feedback – The system will collect user feedback, navigation results, and performance data to inform future improvements.

Development Approach

For the CRIMSON MAP project, the team adopted the Agile software development methodology to ensure a flexible and iterative approach during the system's creation. Agile was chosen because CRIMSON MAP involves developing complex navigation features, all of which require constant adjustments based on real-world testing and user feedback. This also supports fast development, regular feedback, and continuous improvements. Agile is useful when requirements may change or improve over time based on testing or user input. The system also needs to be accurate and adaptable to campus conditions, Agile allows the team to build the system in stages, test its features in actual campus environments, and improve them based on what works best for WMSU students and visitors.

Agile works by breaking the development process into short, repeatable cycles called sprints, where each sprint involves planning, development, testing, and review. At the end of every sprint, the team delivers a working part of the system and gathers feedback to guide the next sprint. This step-by-step development ensures that core

features are tested early and improved continuously, instead of waiting until the end of the project.

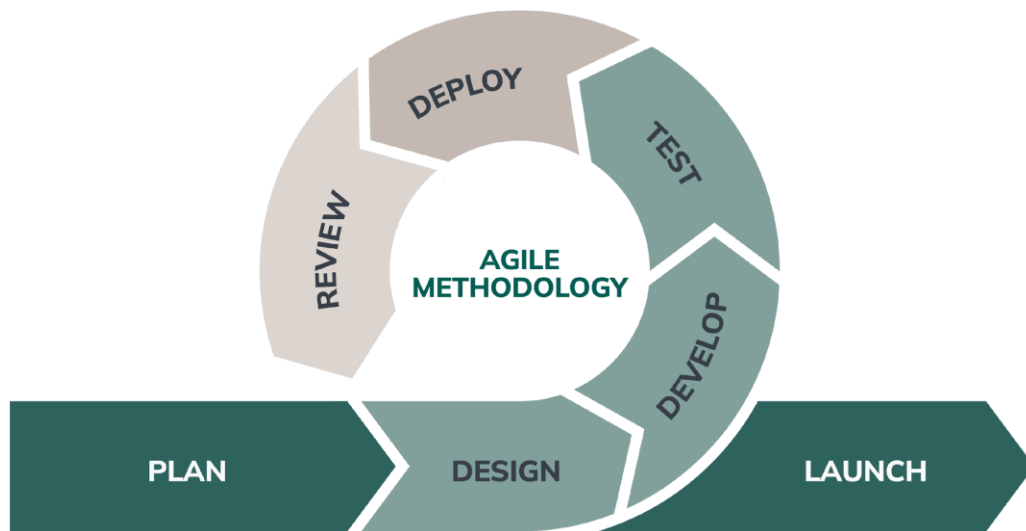


Figure 6: Agile Methodology

The diagram shown in figure 6 illustrates the Agile development method, which is a step-by-step and repetitive process used in software development. It begins with the Planning phase, where the team sets goals and assigns tasks. Then, in the Design phase, the system's layout and interface are prepared. Next, is the Development phase, where the coding and feature implementation are done. After that, the system goes through the Testing phase to identify and fix any issues. Once tested, the system enters the Delivery phase, where the working version is shown to users for feedback. This feedback is then used to improve the system in the next iteration or sprint, repeating the cycle until the final version is complete. This process allows the team to build the system in small parts, test regularly, and make improvements along the way.

System Design

Process Flow Diagram

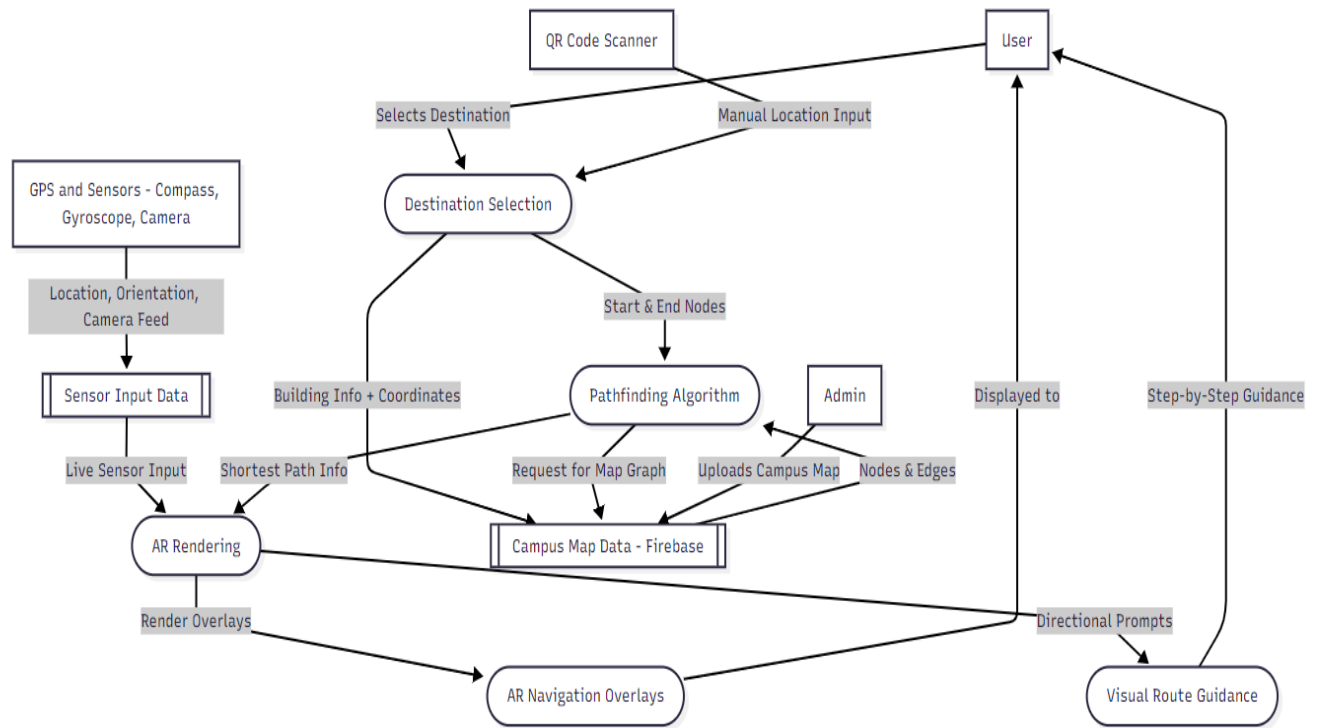


Figure 7 : Data Flow Diagram

Figure 7 presents the Data Flow Diagram of CRIMSON MAP, illustrating how the system processes various user inputs to deliver real-time augmented reality-based navigation within the WMSU campus. Users may scan QR codes placed around the campus to establish their current position. When operating in GPS mode, the system first verifies device GPS compatibility and, once supported, automatically retrieves real-time location and orientation data through GPS and onboard sensors such as the gyroscope, compass, and camera.

These inputs are processed by a pathfinding algorithm that determines the shortest route using campus map data stored in Firebase. The AR module then overlays directional markers and visual prompts onto the camera display, providing step-by-step navigation guidance. Meanwhile, an administrator manages and updates the campus map information within Firebase to maintain accurate and current navigation data. This workflow enables CRIMSON MAP to provide flexible, intuitive, and accessible guidance for students and visitors throughout the campus.

Flowchart

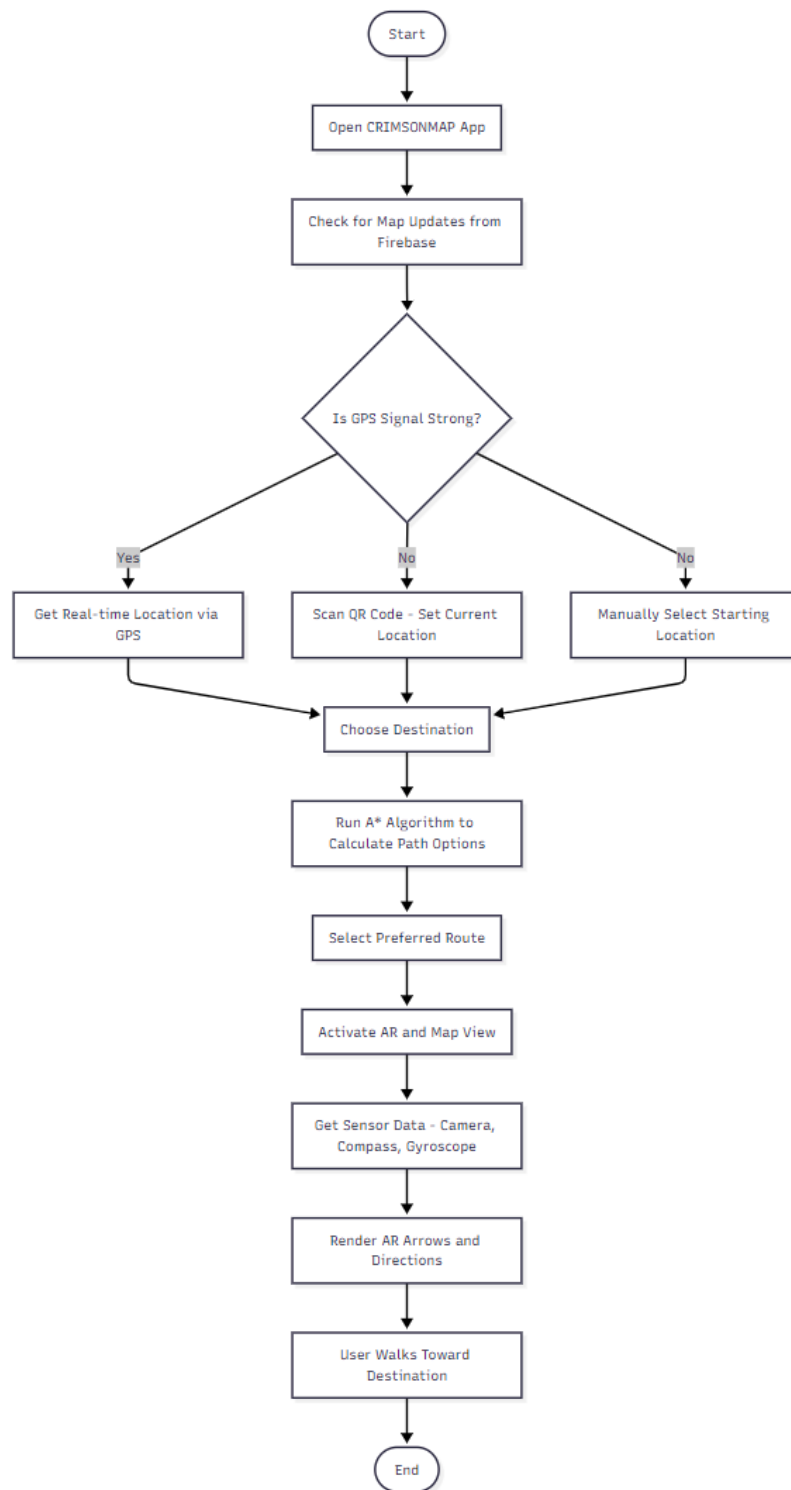


Figure 8: Flowchart

Figure 8 illustrates how the CRIMSON MAP app navigates users throughout the WMSU campus. After the user opens the app, it checks for any map updates from Firebase. It then evaluates the device's GPS signal strength. If GPS is strong, the app automatically retrieves the user's current location. If not, the user has an option: To scan a QR code placed around campus to automatically set their starting point. Once a destination is selected, the app uses the A* algorithm to calculate route options. After selecting a preferred path, the system activates the AR and map view, collects sensor data from the device (camera, compass, gyroscope), and renders visual navigation markers on-screen. These interactive AR overlays help guide users effectively and intuitively to their destination, whether online or offline.

Entity-Relationship Diagram

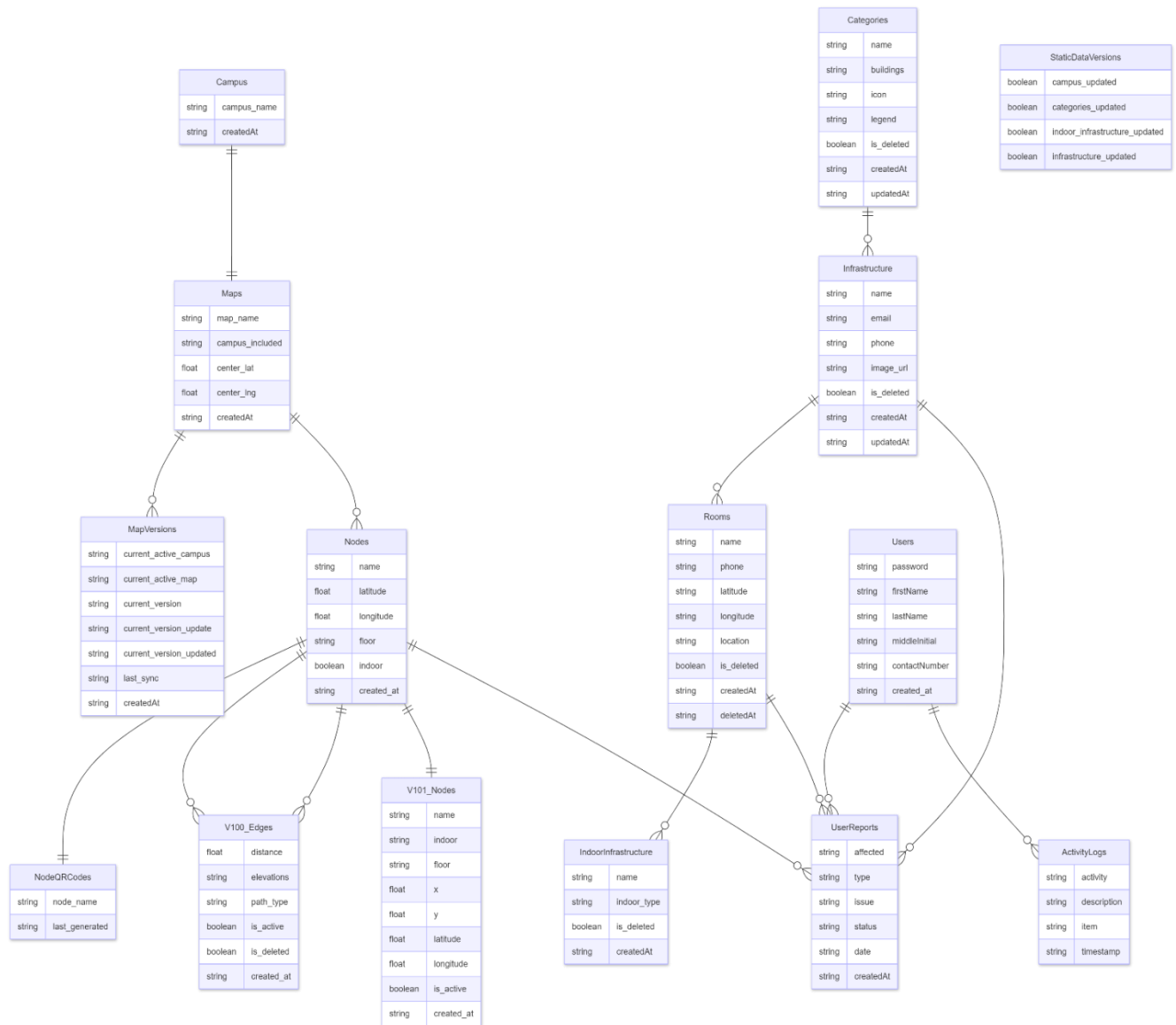


Figure 9: Entity-Relationship Diagram

As shown in figure 9, the data structure of CRIMSON MAP, the system that supports navigation and location services across the WMSU campus. Buildings store detailed information about campus structures and are linked to Paths, Nodes, Edges, User Reports, Activity Logs, and Saved Locations, allowing multiple records for each building.

Paths, along with the Nodes and Edges tables, define the navigation graph used for route computation. Categories classify buildings by type, such as academic or administrative. This design ensures organized, scalable data management, enabling accurate routing, QR code-based location selection, and admin-controlled updates for the mobile navigation app.

System Architecture

The system is designed with a multi-tier architecture to deliver intuitive outdoor navigation across the WMSU campus. At its core, a Firebase Realtime Database serves as the centralized backend, storing essential data such as building coordinates, paths, distance values, and QR code mappings used for location identification. These data points are gathered through coordinate-based mapping and verified using tools like GPS and Google Maps. They are structured into a node-edge graph to support efficient path computation.

The Application Layer handles the core logic, including pathfinding through the A* algorithm and sensor integration. Users can navigate through one of these modes: automatically via GPS, or by scanning a QR code placed at designated areas on campus. These options ensure accessibility even without internet connectivity. All three input methods provide starting points for route calculations, with the shortest path then computed and prepared for rendering.

On the Presentation Layer, the front-end is developed using Unity and ARCore. This interface enables users to search for buildings, choose destinations, and select their preferred navigation method. The mobile device's camera, GPS, and motion sensors, along with ARCore, work together to display real-time visual guidance in the form

of directional markers and overlays. QR codes help enhance positioning accuracy by anchoring user location to pre-registered campus nodes.

Administrative updates to the database, such as building data or QR code assignments, are handled through a web-based panel, allowing staff to keep the system content current. To ensure performance, the architecture includes mechanisms for regular evaluation, testing GPS reliability, QR scan accuracy, AR alignment, and device compatibility.

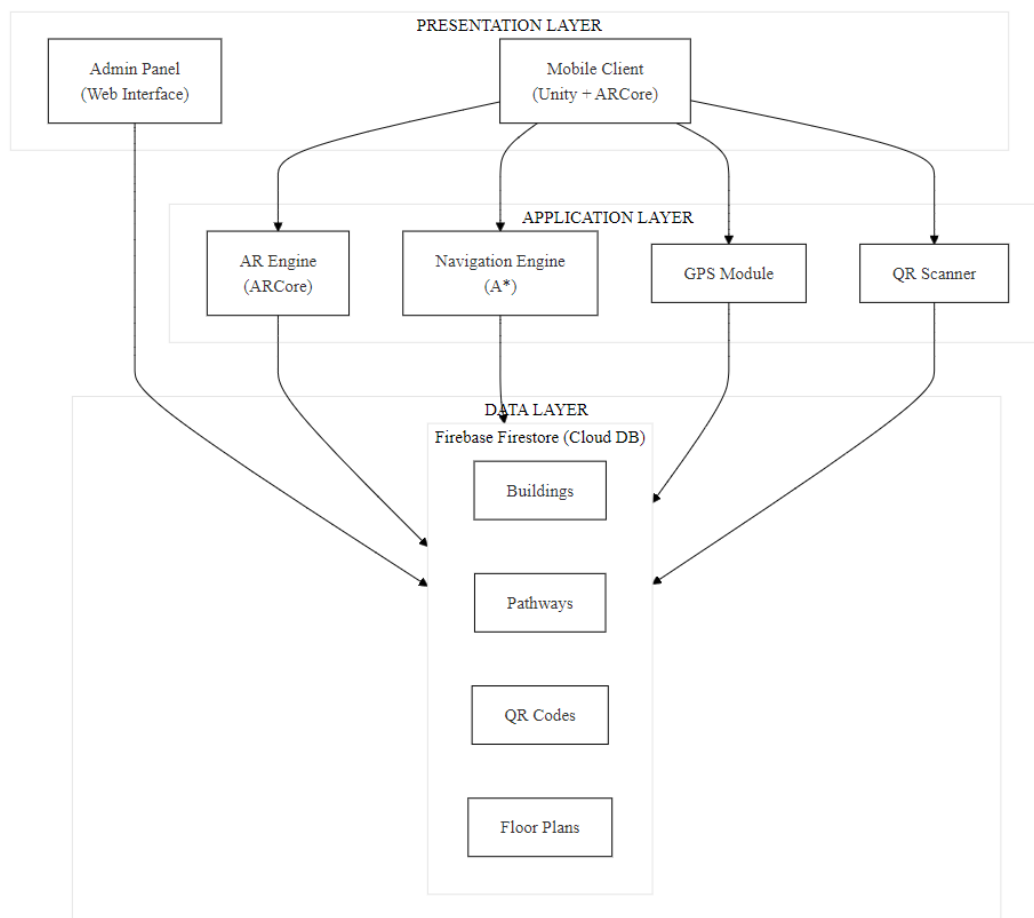


Figure 10: System Architecture

Figure 10 shows how the navigation app works across three layers: the presentation layer, application layer, and data layer. At the top, the presentation layer includes the mobile app

interface where users interact, select destinations, and view AR guidance. In the middle, the application layer handles the logic—like getting sensor data, calculating routes (using A* algorithm), and rendering AR overlays. At the bottom, the data layer stores the map, paths, and QR code information in Firebase. The app supports GPS navigation, making it flexible and accessible for users around the WMSU campus.

Functional Requirements

This section describes the main features and capabilities that the CRIMSON MAP system must perform to support navigation inside Western Mindanao State University. It focuses on what the system must allow the user to do, as well as what the system should handle internally to deliver those functions. These requirements are based on the project's objectives, technical research, and expected user behavior.

The CRIMSON MAP system must support the following core functions:

- Dual AR Modes: Allow users to choose between two AR modes
- Direct AR Mode, which immediately opens the AR scene and displays nearby points of interest (POIs) based on the user's current position.
- AR Navigation Mode, which enables users to navigate from a specific starting point (obtained via GPS or QR code) to a chosen destination using a path-finding algorithm that computes the best route.
- Mode of Localization Selection: Before entering either Direct AR or AR Navigation, prompt users to select their preferred localization mode
- (GPS-based) for real-time positioning, or
- (x–y coordinate system) for areas with weak GPS.
- Recalibration via QR Code: Allow users to scan QR codes anytime while in the AR scene to recalibrate their position

- if GPS signal becomes weak or unstable , or
- if x–y coordinates lose accuracy
- Pathfinding and Route Display: Compute and visualize the shortest and most optimal path between the user’s position and selected destination within the campus.
- Augmented Reality Visualization: Display navigational directions, markers, and information overlays through the device’s camera using AR technology.
- Navigation Audio Prompts: Provide audio prompts that announce the name of a building or infrastructure when the user reaches predefined intermediate points or the destination along the selected route. Audio prompts are triggered based on the user’s position and serve as supplementary guidance to the visual AR interface.
- Digital Map Interface: Provide an interactive map view that complements the AR interface, enabling users to visualize their route and location in 2D.
- Error Reporting and Feedback: Enable users to report incorrect data or suggest map improvements directly through the system interface.

These functions are essential to make sure the app provides accurate, fast, and easy-to-use campus navigation, especially for students and visitors who are unfamiliar with the campus layout.

Testing and Evaluation

To ensure that CRIMSON MAP is accurate, reliable, and user-friendly, the system will undergo systematic testing and evaluation, focusing on both technical performance and user experience. The evaluation follows iterative cycles based on Agile principles and the Design and Development Research (DDR) approach. Each iteration will include testing, feedback collection, and refinements until the system meets the required quality standards.

1. Testing Strategy

- Unit Testing: Each module—including GPS integration, QR code scanning, pathfinding algorithm, AR overlays, and indoor floor plan activation—will be tested individually to ensure proper functionality.
- Integration Testing: All modules will be combined to validate synchronization between Unity, Firebase, GPS, ARCore, and the admin panel.
- System Testing: The complete application will be deployed on-site at WMSU Campus A, B and C. Test paths will include commonly used outdoor routes, indoor building layouts, entrances, and multi-floor areas. Scenarios will include:
 - Recalibrating position via QR codes when GPS is weak.
 - Activating indoor floor plans for building navigation.
 - Admin panel updates and data synchronization testing.

2. Evaluation Criteria

- Accuracy:

GPS and QR-based locations will be compared with physical landmarks for positional accuracy.

Pathfinding will be verified by comparing system-generated routes with actual walking paths on campus.

Indoor floor plan guidance will be assessed for correct room-level positioning.

Target: Locations and routes should closely match real-world measurements, with a mean error within an acceptable threshold

- Efficiency:

App response times, including AR overlay updates, route recalculation, and indoor floor plan rendering, will be measured.

System stability, frame rate, and AR responsiveness will be monitored across multiple Android devices (at least 5 devices with mid-range and high-end specifications).

OS Versions: Android 7.0 (API level 24) and higher.

- Usability:

User surveys will be conducted with students and staff, covering:

- Ease of use in GPS and QR-based modes.
 - Clarity of AR directional markers for outdoor and indoor navigation.
 - Effectiveness of recalibration via QR codes.
 - Overall satisfaction and confidence in following the suggested paths.
 - Observations will note challenges, such as scanning QR codes, switching navigation modes, or using indoor maps.
- Admin Functionality:
 - Authorized staff will test the web-based admin panel for updating buildings, connecting routes, and synchronizing data to the mobile app.
 - Evaluation will check that changes reflect correctly in the AR navigation system without errors.

3. Statistical Validation

Collected data will be analyzed statistically to determine system effectiveness:

- Mean error and standard deviation will be calculated to quantify location and pathfinding precision.

- Hypothesis testing will assess whether CRIMSON MAP offers significant improvements compared to traditional navigation methods (e.g., manual directions or Google Maps).
- Frequency counts and percentages from survey responses will summarize usability and satisfaction levels.

4. Benchmarking and Continuous Improvement

- The performance of CRIMSON MAP will be benchmarked against existing campus navigation solutions in terms of accuracy, efficiency, usability, and accessibility features such as audio guidance.
- Results from testing and user feedback will inform successive Agile iterations, enabling continuous improvements in AR stability, navigation accuracy, system responsiveness, and audio prompt reliability.

CHAPTER IV

RESULTS AND DISCUSSION

Introduction

This chapter presents the results and discussion of the performance evaluation and user assessment of the CRIMSON MAP system. The evaluation is based on systematic testing and user feedback conducted to determine the accuracy, responsiveness, and overall effectiveness of the application in supporting campus navigation through Augmented Reality–assisted outdoor guidance and indoor floor plan viewing.

Performance Evaluation

The results are organized according to key performance metrics, including location accuracy, route correctness, AR loading time, navigation responsiveness, and user satisfaction. Quantitative data gathered from system testing and survey responses are presented using tables and figures to ensure clarity and ease of interpretation.

Each set of results is followed by a corresponding discussion that explains the observed outcomes, identifies contributing factors, and relates the findings to the objectives of the CRIMSON MAP system. The discussion emphasizes how the system performs in real campus environments, drawing insights from both technical measurements and user experiences. Limitations encountered during testing, such as environmental conditions and device-related constraints, are also discussed to provide a realistic and balanced evaluation of the system's performance.

The thirty (n = 30) participants were freshmen students selected from the three WMSU campuses to ensure representation across different colleges. For Campus A, 5 students were chosen from the College of Teacher Education (CTE), 3 from the College of Engineering (COE), and 2 from the College of Liberal Arts (CLA). For Campus B, 4 participants were from the College of Computing Studies (CCS), 4 from the College of Social Work and Community Development (CSWCD), and 2 from the College of Criminal Justice Education (CCJE). For Campus C, 5 students were from the College of Agriculture and 5 from the College of Forestry and Environmental Studies (CFES). The colleges and participant numbers were deliberately chosen to provide balanced input across academic programs and campus locations.

Performance Metrics of CRIMSON MAP Based on 30 Participants (n = 30)

Category	Metric	Average Score (Out of 5)
Functionality (7 items)	GPS Accuracy	4
	Navigation Functions	4.63
	Outdoor Route Accuracy	4.23
	AR Guidance	4.33
	Audio prompts	4.53
	QR Recalibration	4.33
	Meets Navigation Needs	4.40
Indoor Floor Plan (1 item)	Indoor Floor Plan Display	4.80
Reliability (5 items)	No Crashes	4.67
	Multiple Sessions	4.53
	Consistent Results	4.50
	Data Integrity	4.43
	GPS Recovery	4.53
Usability (9 items)	Easy to learn	4.63
	UI-Friendly	4.77
	Visual Appeal	4.77
	Map Switch	4.70
	Find Best Path	4.63
	Efficiency During Navigation	4.57
	Clear Instructions	4.57
	Audio prompt clarity	4.53
	Map Organized	4.57

Efficiency (5 items)	Fast Path Calculation	4.47
	Fast Response	4.23
	No Lag	4.67
	QR Accurate	4.57
	Time Saving	4.53
Overall Satisfaction (4 items)	Meets Purpose	4.60
	Enhances Experience	4.50
	Satisfactory Performance	4.60
	Recommend	4.53

Table 4.1: Performance Metrics

Table 4.1 presents the average scores of the CRIMSON MAP system based on the evaluations of thirty (n = 30) participants. The assessment used a system evaluation form that measured the application's performance across six major categories: Functionality, Indoor Floor Plan, Reliability, Usability, Efficiency, and Overall Satisfaction. All indicators were rated using a 5-point Likert scale, where 5 means Strongly Agree and 1 means Strongly Disagree.

In Functionality, mean scores ranged from 4.00 to 4.63, showing that participants generally agreed that the navigation features are effective. GPS Accuracy received the lowest score (4.00), while Navigation Functions and Audio Prompts scored the highest. The Indoor Floor Plan feature obtained a high score of 4.80, indicating its usefulness for indoor navigation.

All Reliability metrics scored above 4.43, demonstrating that the system is stable, consistent, and operates without crashes. Usability ratings ranged from 4.53 to 4.77, showing that the interface is easy to use, visually appealing, and well-organized. Efficiency scores (4.23–4.67) reflected fast calculations, accurate QR scanning, and minimal lag. Finally, Overall Satisfaction scores ranged from 4.50 to 4.60, indicating that participants found the system effective, helpful, and recommendable.

Overall, the results indicate that CRIMSON MAP is a reliable, efficient, and user-friendly AR-based navigation system.

Objective 1: Application Design and AR Integration

The CRIMSON MAP mobile application was developed using Unity together with ARCore to enable augmented reality–based campus navigation. Unity served as the main development environment for system architecture and interface design, while ARCore provided environmental tracking and rendering of virtual navigation elements. Through these technologies, directional markers, route indicators, and points of interest are superimposed onto the real-world campus environment.

The application offers two AR interaction modes. Direct AR Mode allows users to immediately view nearby points of interest based on their current location, while AR Navigation Mode provides guided navigation from a selected starting point to a chosen destination. Additional supporting features include AR visual overlays, a digital map interface, QR-code recalibration, and audio navigation prompts, which together strengthen the overall AR integration and user experience.

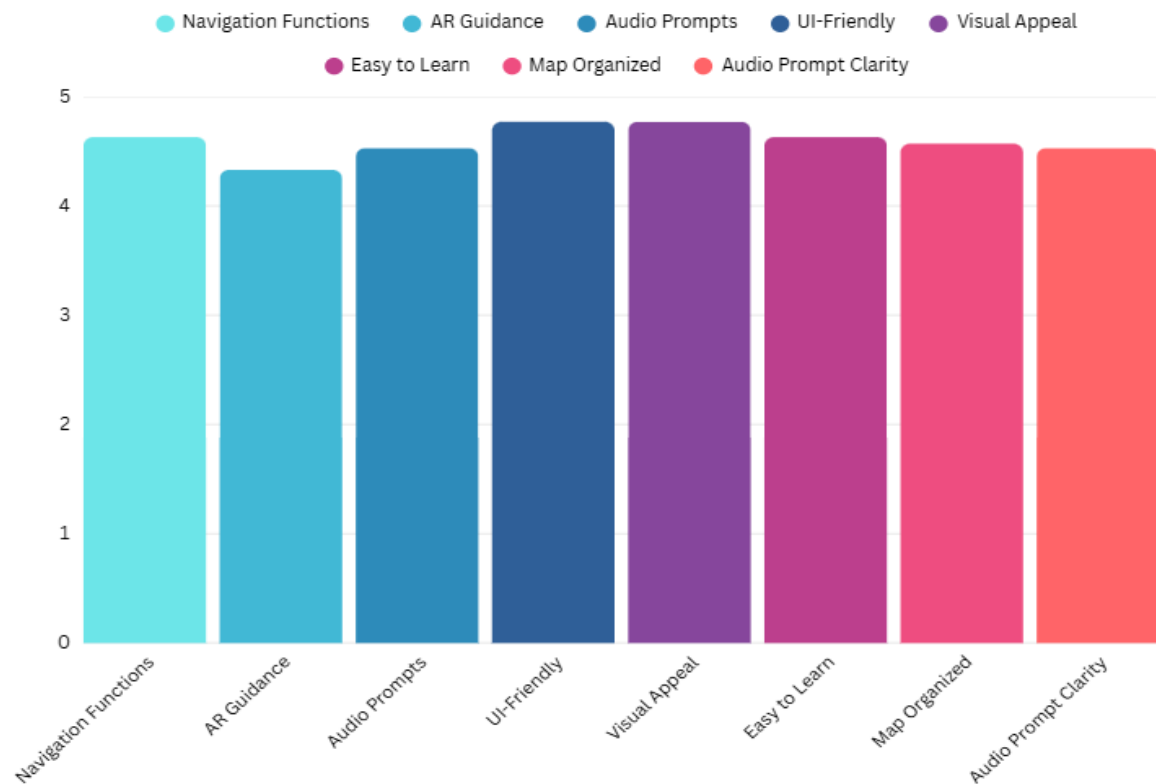


Figure 11: Mean scores of AR-related functionality and usability indicators

Figure 11 shows the evaluation results from 30 participants ($n = 30$) show strong performance across both functionality and usability measures related to AR integration. Navigation Functions achieved a mean score of 4.63, and AR Guidance obtained 4.33. Audio Prompts received 4.53, suggesting that the auditory cues effectively complemented the visual AR guidance.

From a usability perspective, the application interface received particularly high ratings. UI-Friendly and Visual Appeal both reached mean scores of 4.77, indicating a high level of user satisfaction with the interface design. Easy to Learn scored 4.63, while Map Organized and Audio Prompt Clarity obtained 4.57 and 4.53, respectively. These findings

demonstrate that the AR-based design is intuitive, visually appealing, and suitable for campus navigation.

Objective 2: GPS-based Navigation

GPS localization was implemented as the primary positioning method for outdoor navigation within the CRIMSON MAP application. The system continuously monitors the user's location and updates the navigation path, enabling users to move efficiently toward selected destinations across the campus.

This functionality is supported by GPS tracking, outdoor route computation, and continuous path updates. The A* pathfinding algorithm determines the optimal route between the user's current position and the selected destination. When GPS signals weaken, the QR code recalibration feature allows users to re-establish accurate positioning and continue navigation.

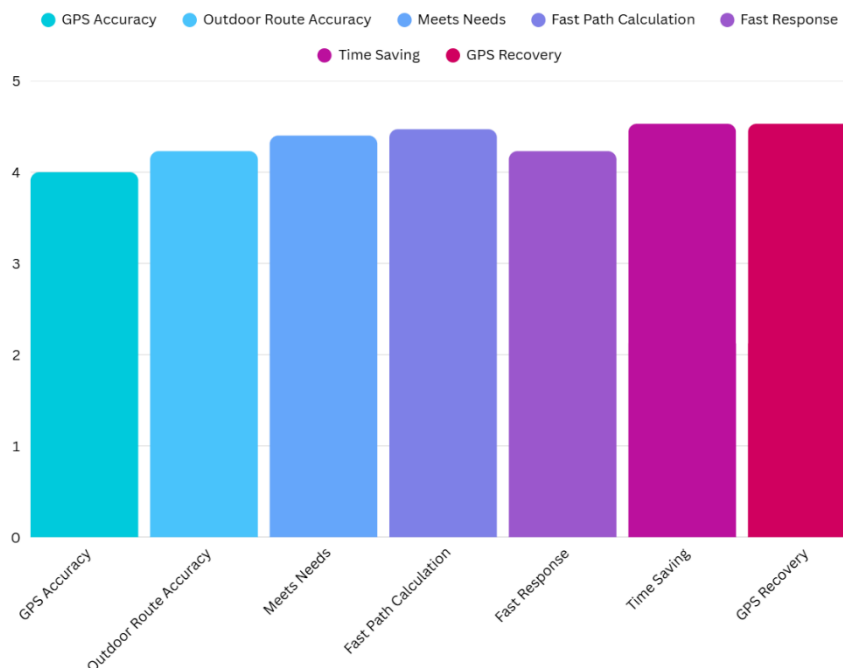


Figure 12: Mean scores of GPS-based navigation performance indicators

Figure 12 presents the evaluation conducted with 30 participants ($n = 30$) indicates positive results across functionality, efficiency, and reliability metrics. GPS Accuracy received a mean score of 4.00, while Outdoor Route Accuracy obtained 4.23. Meets Navigation Needs recorded 4.40, showing that users considered the GPS navigation effective for outdoor routing.

Efficiency-related indicators also yielded strong ratings. Fast Path Calculation achieved 4.47, Fast Response obtained 4.23, and Time Saving recorded 4.53, highlighting the system's responsiveness and computational efficiency. GPS Recovery reached a mean score of 4.53, reflecting reliable restoration of positioning during signal disruptions. Overall, these findings confirm the successful implementation of GPS-based navigation.

Objective 3: Navigation Audio Support

This objective was achieved by integrating audio-based navigation prompts into the CRIMSON MAP application. It provides users with verbal guidance to complement the visual AR interface. This feature was designed to improve accessibility and reduce the need for continuous screen interaction during navigation. The system includes audio prompts that announce building names based on the user's location. These prompts operate alongside AR visual markers and the digital map interface, ensuring that users receive clear and timely guidance while navigating the campus.



Figure 13: Mean scores of audio navigation-related indicators

Results from the evaluation involving 30 participants ($n = 30$) indicate strong acceptance of the audio navigation feature across functionality, usability, and overall satisfaction measures as shown in Figure 13. Audio Prompts obtained a mean score of 4.53, while Audio Prompt Clarity also recorded 4.53, showing that participants considered the audio guidance clear and effective.

Regarding usability, Easy to Learn achieved a mean score of 4.63, suggesting that users were able to quickly understand and use the audio feature. Under overall satisfaction, Enhances Experience recorded 4.50, indicating that the audio guidance positively contributed to the overall navigation experience. These findings confirm the successful implementation of navigation audio support.

Objective 4: Pathfinding

This objective was achieved by implementing the A* pathfinding algorithm within the CRIMSON MAP application. The algorithm calculates the optimal route between the user's current location and the selected destination by considering distance and predefined campus paths.

The system supports route computation using the A* algorithm, allowing users to select a destination and receive a visual path in both AR and map modes.

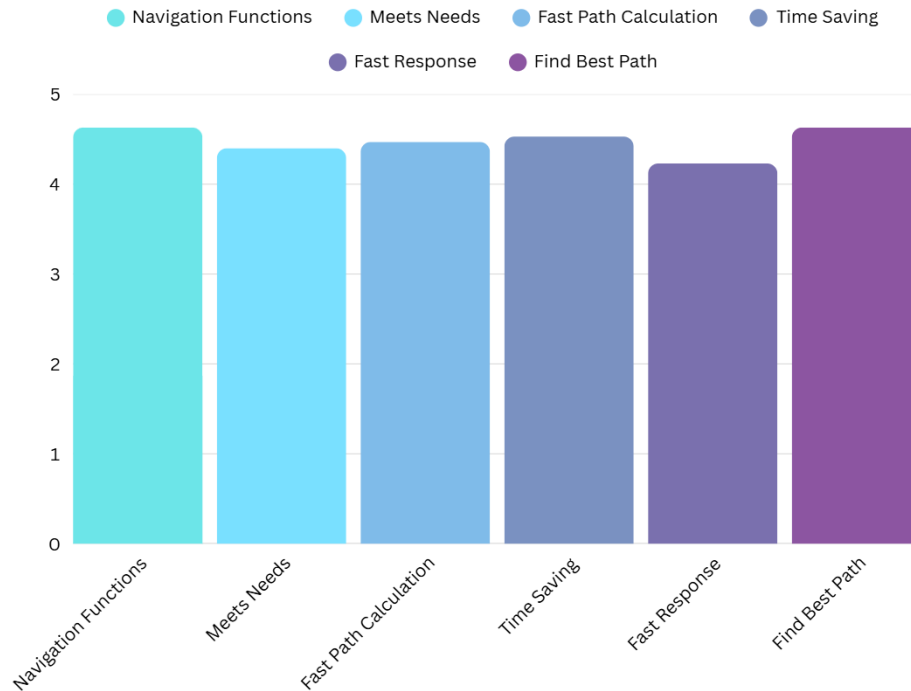


Figure 14: Mean scores of pathfinding-related performance indicators

Figure 14 shows the responses of 30 participants ($n = 30$), the pathfinding functionality received consistently high ratings across functionality, efficiency, and usability indicators. Navigation Functions achieved a mean score of 4.63, while Meets Navigation Needs recorded 4.40, demonstrating that users found the routing capability effective for campus navigation.

Efficiency metrics also showed positive results. Fast Path Calculation obtained 4.47, Time Saving reached 4.53, and Fast Response recorded 4.23, indicating fast and responsive route computation. From a usability standpoint, Find Best Path achieved a mean score of 4.63, confirming that participants perceived the system as capable of generating optimal routes. These results indicate that the pathfinding objective was successfully met.

Objective 5: Recalibration via QR Codes

To ensure continuous navigation despite positioning errors, a QR code–based recalibration feature was incorporated into the CRIMSON MAP application. This mechanism allows users to restore accurate location data whenever GPS positioning becomes unreliable. After scanning a QR code placed at strategic campus locations, the system updates the user’s position and recalculates the route to the selected destination.

This feature operates in coordination with the navigation engine to minimize disruption during movement. Efficient route recalculation and responsive system behavior help users resume navigation quickly and maintain a smooth navigation experience.

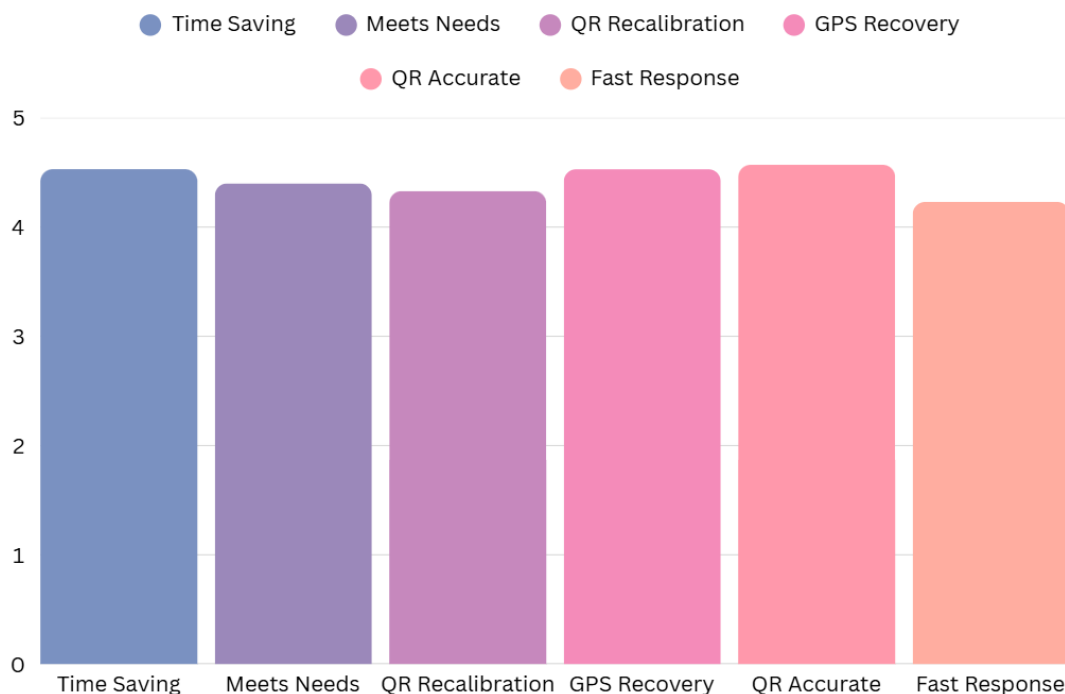


Figure 15: Mean scores of QR recalibration–related indicators

Evaluation results from 30 participants (n = 30) show strong agreement regarding the effectiveness of the recalibration feature shown in figure 15. QR Accuracy obtained the highest mean score of 4.57, while Time Saving and GPS Recovery both achieved 4.53, indicating that users were able to restore accurate positioning and continue navigation efficiently.

Meets Needs recorded a mean score of 4.40, suggesting that participants considered the recalibration capability sufficient for campus navigation. QR Recalibration achieved 4.33, and Fast Response recorded 4.23. Although Fast Response received the lowest score among the indicators, it still reflects positive user agreement. Overall, these findings demonstrate that QR-based recalibration effectively supports reliable and uninterrupted navigation.

Objective 6: Indoor Floor Plan Integration

This objective was achieved by integrating an indoor floor plan viewing feature into the CRIMSON MAP application. When a user scans a building that has an available indoor layout, an Indoor button becomes visible, allowing the user to switch from the AR navigation interface to a 2D indoor map view. The feature enables users to view building layouts and identify key rooms and areas without providing real-time indoor tracking or turn-by-turn directions.

The system provides a dedicated indoor floor plan display that presents the building layout in a clear and organized manner. The indoor map is accessible only for

buildings with available indoor data. This feature complements the AR-based outdoor navigation by offering visual reference once users enter a building.

The indoor floor plan feature was evaluated using a single indicator, Indoor Floor Plan Display, based on responses from 30 participants ($n = 30$). The results show a high mean score of 4.80, indicating strong user agreement that the indoor floor plan was clear, understandable, and useful.

The consistently high ratings suggest that participants found the indoor map effective for viewing building layouts and locating key rooms, even without real-time indoor navigation. These findings confirm that the objective of integrating an indoor floor plan viewing feature was successfully met.

Objective 7: Admin Panel Development

This objective was accomplished by developing a web-based administrative panel that allows authorized campus personnel to manage and update campus navigation data used by the CRIMSON MAP mobile application. The admin panel was deployed online to enable real-time synchronization between the web system and the mobile application.

The system was designed to allow administrators to easily maintain and update campus infrastructure data, ensuring that the navigation information remains accurate and up to date. The admin panel provides a structured dashboard and sidebar navigation that allows administrators to access multiple management modules. The dashboard displays an overview of the system, including recent activity logs, the total number of buildings,

rooms, and offices, and a current map overview. The Manage Buildings module allows administrators to add and update campus data, including Infrastructure, Indoor Infrastructure, Categories, Maps, and Campus information. A Map Editor module enables editing of map details such as latitude and longitude coordinates, infrastructure connections, and whether a location is indoor or outdoor. A Recent Activity page serves as a history log that records updates made by administrators. The Reports module allows administrators to view user-submitted reports about blocked routes, closed paths, flooding, and other navigation issues, categorized as nodes or edges. The system also includes an Account page for administrator profile management and a Logout function for secure access control. All updates made in the admin panel are synchronized with the mobile application to ensure consistency between the system database and the user interface.

System Testing Results

The admin panel was tested by the researchers through continuous system use and data updates. The results showed that the panel successfully supports adding, editing, and synchronizing campus navigation data with the mobile application. Changes made through the admin interface were reflected correctly within the mobile app, confirming successful integration between the web and mobile components of the system.

During testing, the admin panel performed reliably across its modules, including building management, map editing, activity tracking, and report monitoring. The only observed limitation was that page loading time may increase when internet connectivity is unstable, indicating that the system performs best under a stable internet connection.

Summary of Objective Achievement

Overall, the results presented in this chapter demonstrate that all objectives of the CRIMSON MAP project were successfully achieved. The developed mobile application effectively integrates Augmented Reality, GPS-based navigation, audio guidance, pathfinding, QR-based recalibration, and indoor floor plan viewing to support efficient campus navigation. Evaluation results from 30 participants consistently produced high mean scores across functionality, usability, efficiency, reliability, and overall satisfaction indicators, confirming that the system is accurate, responsive, easy to use, and beneficial for campus users. In addition, system testing of the web-based admin panel verified that campus infrastructure data can be efficiently managed and synchronized with the mobile application. The combined results of user evaluation and system testing confirm that the CRIMSON MAP system meets its intended purpose of providing a reliable and user-friendly campus navigation solution for Western Mindanao State University.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

This chapter presents the conclusion and recommendations derived from the findings and analyses discussed in the previous chapter. It summarizes the key outcomes of the study and highlights the contributions of the developed system. Furthermore, it provides practical recommendations intended to guide future improvements and related research on the CRIMSON MAP application.

Conclusion

This study successfully designed, developed, and evaluated CRIMSON MAP, an Augmented Reality–based campus navigation mobile application for Western Mindanao State University Main Campus. It is therefore concluded that:

- Objective 1: Application Design and AR Integration

The CRIMSON MAP mobile application was successfully developed using Unity and ARCore, providing an intuitive AR-based navigation experience. The integration of directional markers, route indicators, and points of interest into the real-world environment was effectively achieved. Evaluation results show high ratings in AR guidance, navigation functions, and usability, confirming that the application design supports effective AR-based campus navigation.

- Objective 2: GPS-Based Navigation

The system successfully implemented GPS as the primary method for outdoor localization, enabling accurate positioning and continuous route updates under normal GPS conditions. User evaluation indicated positive performance in

GPS accuracy, outdoor route correctness, and navigation efficiency, confirming that the GPS-based navigation reliably guides users across campus.

- Objective 3: Audio-Based Navigation Support

Audio prompts were effectively integrated into CRIMSON MAP to provide verbal guidance alongside visual AR cues. Participants rated the audio navigation highly for clarity and usability, indicating that the feature enhances accessibility and reduces reliance on continuous screen interaction during navigation.

- Objective 4: Pathfinding Algorithm

The A* pathfinding algorithm successfully calculated optimal routes between the user's current location and selected destinations. Evaluation results show high scores in navigation effectiveness, fast path computation, and time-saving, confirming that the pathfinding functionality supports efficient and reliable campus navigation.

- Objective 5: QR Code Recalibration

The QR code-based recalibration feature effectively restored accurate location data when GPS signals were weak or positional errors occurred. Participants rated QR accuracy, GPS recovery, and time-saving positively, demonstrating that recalibration supports uninterrupted navigation and enhances user confidence in the system.

- Objective 6: Indoor Floor Plan Integration

The indoor map feature allowed users to view building layouts and identify key rooms through an Indoor button when a floor plan was available. Evaluation results showed a high mean score for clarity and usefulness, confirming that indoor floor plan integration complements AR-based outdoor navigation and assists users inside campus buildings.

- Objective 7: Admin Panel Development

The web-based administrative panel was successfully developed to allow authorized campus personnel to manage, update, and synchronize campus navigation data with the mobile application. System testing confirmed that administrators could reliably maintain building data, edit maps, track activities, and monitor user reports. This ensures that the mobile application reflects accurate and up-to-date campus navigation information.

Recommendations

Based on the findings, user feedback, and observed system limitations, the following recommendations are proposed for future development and improvement of CRIMSON MAP.

A. Technical Improvements

1. Enhance GPS Accuracy

Implement sensor fusion techniques combining GPS, compass, and inertial sensors to reduce outdoor positioning drift in weak-signal areas.

2. Improve AR Stability

Further optimize AR rendering and tracking to improve performance on mid-range and lower-end Android devices.

3. Expand QR Recalibration Coverage

Place additional QR anchors in areas where GPS signals are commonly unstable to improve recalibration speed and accuracy.

4. Optimize Admin Panel Performance

Improve page loading performance and caching mechanisms to reduce dependency on strong internet connectivity.

B. Feature Enhancements

5. Add Accessibility Features

Introduce high-contrast mode, larger text options, and additional audio feedback to support users with accessibility needs.

6. Implement Push Notifications

Allow users to receive real-time alerts about blocked paths, construction, or emergency situations reported through the admin panel.

C. Platform Expansion

7. Develop an iOS Version

Expanding the application to iOS would increase accessibility and allow more users to benefit from the system.

D. Deployment and Future Research

8. Conduct Large-Scale Deployment Testing

Future studies should involve a larger and more diverse group of users over an extended period.

9. Establish Continuous Data Maintenance

Regular updates of buildings, rooms, and infrastructure should be maintained through coordination with university offices.

References

- [1] F. Lu, H. Zhou, L. Guo, J. Chen, and L. Pei, "An ARCore-based augmented reality campus navigation system," *Applied Sciences*, vol. 11, no. 16, p. 7515, Aug. 2021. doi: [10.3390/app11167515](https://doi.org/10.3390/app11167515)
- [2] C.-H. Lin, Y. Chung, B.-Y. Chou, H.-Y. Chen, and C.-Y. Tsai, "A novel campus navigation app with augmented reality and deep learning," in *Proc. IEEE Int. Conf. Applied System Invention (ICASI)*, Chiba, Japan, Apr. 2018, pp. 1075–1077. doi: [10.1109/ICASI.2018.8394464](https://doi.org/10.1109/ICASI.2018.8394464)
- [3] D. Croce, L. Giarre, F. Pascucci, I. Tinnirello, and A. L. Valvo, "An indoor and outdoor navigation system for visually impaired people," *IEEE Access*, vol. 7, pp. 170406–170418, Nov. 2019. doi: [10.1109/ACCESS.2019.2955046](https://doi.org/10.1109/ACCESS.2019.2955046)
- [4] P. E. Kourouthanassis, C. Boletsis, and G. Lekakos, "Demystifying the design of mobile augmented reality applications," *Multimedia Tools and Applications*, vol. 74, no. 4, pp. 1045–1066, Feb. 2015. doi: [10.1007/s11042-013-1710-7](https://doi.org/10.1007/s11042-013-1710-7)
- [5] P. Verma, K. Agrawal, and V. Sarasvathi, "Indoor navigation using augmented reality," in *Proc. 4th Int. Conf. Virtual Augmented Reality Simulations*, Sydney, Australia, Feb. 2020, pp. 58–63. doi: [10.1145/3385378.3385387](https://doi.org/10.1145/3385378.3385387)
- [6] M. K. Sharma, S. Chachaundiya, and Vishal, "Augmented Reality Navigation," *Int. J. Eng. Res. Technol.*, vol. 9, no. 6, pp. 998–1002, Jun. 2020. doi: [10.17577/IJERTV9IS060441](https://doi.org/10.17577/IJERTV9IS060441)

- [7] S. Graser, F. Kirschenlohr, and S. Böhm, "User Experience Evaluation of Augmented Reality: A Systematic Literature Review," *arXiv*, Nov. 2024. doi: 10.48550/arXiv.2411.12777
- [8] S. Singh, J. Singh, B. Shah, S. S. Sehra, and F. Ali, "Augmented reality and GPS-based resource-efficient navigation system for outdoor environments: Integrating device camera, sensors, and storage," *Sustainability*, vol. 14, no. 19, p. 12720, Oct. 2022. doi: [10.3390/su141912720](https://doi.org/10.3390/su141912720)
- [9] N. El Barhoumi, R. Hajji, Z. Bouali, Y. B. Brahim, and A. Kharroubi, "Assessment of 3D models placement methods in augmented reality," *Applied Sciences*, vol. 12, no. 20, p. 10620, Oct. 2022. doi: [10.3390/app122010620](https://doi.org/10.3390/app122010620)
- [10] M. Bermejo-Martins, F. Gutierrez, and J. M. Sánchez, "Geolocation-based augmented reality with UnityGeoAR," in *Proc. Int. Conf. Geographical Information Systems Theory, Applications and Management (GISTAM)*, 2021, pp. 133–141. doi: 10.5220/0010488301330141
- [11] A. Alsabbagh, S. Alkhalaf, and M. Al-Mashaqbeh, "University campus indoor navigation using AR and A* search algorithm," in *Proc. 12th Int. Conf. Information and Communication Systems (ICICS)*, Valencia, Spain, May 2021, pp. 362–367. doi: 10.1109/ICICS52457.2021.9464583
- [13] C. Ankora and A. D., "Integrating User Stories in the Design of Augmented Reality Application," *International Journal of Information Technologies and Systems Approach*, vol. 15, no. 1, pp. 1–19, Jan. 2022. doi: 10.4018/IJITSA.304809

- [14] Z. An, X. Rui, and C. Gao, "Improved A* Navigation Path-Planning Algorithm Based on Hexagonal Grid," *ISPRS International Journal of Geo-Information*, vol. 13, no. 5, p. 166, May 2024. [Online]. Available: <https://doi.org/10.3390/ijgi13050166>
- [15] S. Singh, J. Singh, B. Shah, S. S. Sehra, and F. Ali, "Augmented Reality and GPS-Based Resource Efficient Navigation System for Outdoor Environments: Integrating Device Camera, Sensors, and Storage," *Sustainability*, vol. 14, no. 19, p. 12720, Oct. 2022. doi: [10.3390/su141912720](https://doi.org/10.3390/su141912720)
- [16] A. Rai, "A study on Bellman Ford algorithm for shortest path detection in Global Positioning System," *Int. J. Recent Advances in Science, Engineering and Technology (IJRASET)*, vol. 10, no. 7, pp. 227–235, Jul. 2022. doi: [10.22214/ijraset.2022.42720](https://doi.org/10.22214/ijraset.2022.42720)
- [17] B. S. C. Putra, I. K. D. Senapatha, J.-C. Wang, M. B. Nendya, D. Pandapotan, F. N. Tjahjono, and H. B. Santoso, "Adaptive AR Navigation: Real-Time Mapping for Indoor Environment Using Node Placement and Marker Localization," *Information*, vol. 16, no. 6, p. 478, Jun. 2025. [Online]. Available: <https://doi.org/10.3390/info16060478>
- [18] Western Mindanao State University, "History," [Online]. Available: https://wmsu.edu.ph/?page_id=313
[Accessed: Oct. 25, 2025].
- [19] Wikipedia contributors, "Western Mindanao State University," *Wikipedia, The Free Encyclopedia*, [Online]. Available: https://en.wikipedia.org/wiki/Western_Mindanao_State_University

[Accessed: Oct. 25, 2025].

[20] R. Hashimoto and M. Cohen, "Outdoor Navigation System by AR," *SHS Web of Conferences*, vol. 102, p. 04002, Jan. 2021. DOI: [10.1051/shsconf/202110204002](https://doi.org/10.1051/shsconf/202110204002)

[21] Oxford University Press, *Oxford Dictionary of Computing*, 8th ed. Oxford, UK: Oxford University Press, 2023.

[22] U.S. Department of Defense, "Official U.S. government information about the Global Positioning System (GPS) and related topics," *GPS.gov*, 2023. [Online]. Available: <https://www.gps.gov/>

[23] Google, "Assisted GPS (A-GPS)," *Android Developers Documentation*, 2024. [Online]. Available: <https://developer.android.com/>

[24] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2021.

[25] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, pp. 269–271, 1959. doi: 10.1007/BF01386390

[26] Techopedia, "Node," 2024. [Online]. Available: <https://www.techopedia.com/>

[27] Techopedia, "Edge (graph theory)," 2024. [Online]. Available: <https://www.techopedia.com/>

[28] Google, "Firebase Documentation," 2024. [Online]. Available: <https://firebase.google.com/docs>

[29] Unity Technologies, "Unity Documentation," 2024. [Online]. Available: <https://docs.unity.com>

[30] Unity Technologies, “AR Foundation Documentation,” 2024. [Online].

Available: <https://docs.unity3d.com/Packages/com.unity.xr.arfoundation>

[31] “Digital Wayfinding in Smart Cities,” *Smart Cities Journal*, vol. 6, no. 2, pp. 45–52,

Appendix A: Data Collected

Appendix D: Relevant Source Code

Appendix E: Ethical Clearance

Appendix F: Plagiarism Report

Appendix G: Research Critique and Editing Certificate

Appendix H: Curriculum Vitae



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An Information Technology student with hands-on experience in coding across multiple platforms, including web design, web development, and application development. Known for meeting deadlines with quality results, and viewing errors as opportunities to gain new knowledge. Highly motivated to learn and adapt, with strong communication skills and a collaborative mindset, while also taking initiative to contribute ideas and deliver work with dedication.

EDUCATION

Primary Education:
Talusán Elementary School, Graduated 2016

Secondary Education:
Talusán National High School,
Graduated 2020

Senior High:
Talusán National High School,
Graduated 2022

Tertiary Education:
Bachelor of Science in Information
Technology (BSIT), 4th Year Student,
Batch 2026

WORK EXPERIENCE

Front-End and Back-End Developer
Application Development (2024)
studya.io

- A student app that combines the Pomodoro method with flashcards to boost focus and learning.

Front-End and Back-End Developer
Web Application Development (2025)
Art de Rio Photography Booking System

- A web-based booking system designed to streamline booking operations.

SKILLS

- Communication
- Collaboration
- Project Management
- Active Listening
- Flexibility

TECHNICAL SKILLS

- HTML
- CSS
- JavaScript
- Flutter
- Python



DARYL T. GREGORIO

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- Baliwasan, Zamboanga City

I am an IT student with a strong passion for creativity, design, and innovation. I enjoy working on projects that involve visual arts, video editing, image editing, and front-end development. With a keen eye for detail and design, I bring both technical and artistic skills to my work. I am adaptable, skilled at meeting timebound tasks, and able to collaborate effectively within teams to achieve shared goals.

EDUCATION

Primary Education:
Bolong Elementary School, Graduated 2016

Secondary Education:
Bolong National High School, Graduated 2020

Senior High School:
Bolong Senior High School, Graduated 2022

Tertiary Education:
Bachelor of Science in Information Technology (BSIT), 4th Year Student, Batch 2026

WORK EXPERIENCE

Video Editor (Part-Time)
Western Mindanao Power Corporation
• 2020 - 2021

Front-End and Back-End Developer
Application Development (2024) - VocabVoyage
• An app that helps users to improve their spelling skills through engaging, level-based challenges in an ocean-themed adventure

VOLUNTEER & LEADERSHIP EXPERIENCE

President - Youth for Environment in Schools Organization (YES-O)
Bolong National High School
2020 - 2021
• Led the organization in promoting environmental awareness and sustainable practices.
• Organized and participated in a tree planting activity, contributing to community greening efforts.

SKILLS

- Multitasking
- Teamwork and Collaboration
- Creative Solutions
- Verbal and written communication
- Time Management
- Flexibility

TECHNICAL SKILLS

- HTML
- CSS
- JavaScript
- Python
- Kotlin



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Motivated IT student with a strong foundation in mobile and web development, specializing in front-end solutions and cross-platform applications. Skilled in Flutter, .NET, JavaScript, and web technologies, with hands-on experience building user-friendly and responsive applications. A quick learner with strong problem-solving skills and a passion for creating innovative digital solutions. In my free time, I enjoy 'telling computers what to do'—sometimes they listen, most of the time they don't, and that's what makes coding exciting

EDUCATION

Bachelor of Science in Information
Technology
Western Mindanao State University,
Expected 2026

Senior High School
Chong Hua High School, 2021-2022

Secondary Education
Talon-Talon National High School,
2016-2020

Primary Education
Talon-Talon Elementary School, 2010-
2016

WORK EXPERIENCE

Video Editor

Freelance / Work-from-Home

February 2022 - November 2022

- Edited YouTube content focused on financial literacy, online income strategies, and tool tutorials.
- Produced engaging video cuts, transitions, and effects to maintain audience interest.

Front-End and Back-End Developer

Application Development (2024) - studya.io

- A student app that combines the Pomodoro method with flashcards to boost focus and learning.

AI-Furqan Enrollment and Academic System (2025)

- Built a web-based academic and enrollment system to manage student registration, payments, sectioning, and grade records.
- Developed the front-end interface using HTML, CSS, and JavaScript, focusing on responsive and user-friendly design.

SKILLS

- Problem-solving & critical thinking
- Team collaboration & communication
- Adaptability & continuous learning
- Attention to detail

TECHNICAL SKILLS

- Mobile Development: Flutter, Firebase, Unity
- Web Development: HTML, CSS, JavaScript, .NET (C#, VB.Net)
- Front-End Development: Responsive design, UI/UX best practices
- Tools & Platforms: Git/GitHub, Visual Studio, VS Code